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# Land Cover Dynamics and Transitions in Italy: A Multi-Temporal Analysis Using Global Land Cover Datasets

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# Dedication

I couldn't have walked this long road alone. This work is a testament to the incredible people who walked beside me.

To my wife Najibeh Movahedi. You were my anchor when I felt completely lost. Thank you for every word of encouragement, for every sacrifice you made without me even asking, and for believing in me even on the days I didn't believe in myself. This achievement is ours, and I share it with you completely.

To Stefano Pasta, my brother. You were my first welcome in a new country and you became family. You shared your home, your wisdom, and your friendship, and you gave me the courage to keep going when everything felt unfamiliar and difficult. I will never forget what you've done for me.

And finally, to the women and girls of Afghanistan. In the face of unimaginable adversity, your resilience and quiet strength are a powerful reminder of why the pursuit of knowledge matters. This work is dedicated to your fight and to the hope that doors will open for you once again.

To all of you, my deepest thanks. You carried me here.

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# Abstract

Land abandonment and subsequent ecological transitions are key processes influencing the Italian landscape, driven by socio-economic and environmental factors. This research provides a national-scale assessment of these dynamics from 1985 to 2032, using two global land cover datasets: the ESA CCI Land Cover (300 m resolution, 1992–2022) and the GLC_FCS30D (30 m resolution, 1985–2022). The analysis was implemented within a GRASS GIS environment with automated scripting to ensure consistency and reproducibility.

The methodology was organized around four components: (1) pixel-based statistical analysis quantifying the annual extent of each land cover class, with dynamic pixel-area calculation for cartometric accuracy; (2) computation of transition matrices to identify conversion pathways, particularly from agricultural to natural land covers; (3) comparative validation of the two datasets, resampling the 30 m data to 300 m to assess consistency through confusion matrices and accuracy metrics, showing moderate overall agreement (78–79%) with poor result for heterogeneous classes such as Shrubland (User's Accuracy ≈25%), and higher consistency for stable classes such as Water (>88%) and Cropland (>88%); and (4) development of a constrained Markov Chain model, integrating Italy-specific ecological and socio-economic constraints, to project land cover to 2032.

The results indicate a landscape characterized by both persistence and change. In 2022, cropland accounted for approximately 48% of national territory and forest for 36%. Built-up areas expanded by 70–104% depending on dataset, representing the most significant transformation between 1985 and 2022. Forest cover showed a slight net decrease (−2.6%), while grassland declined more substantially (−15.2%). Projections to 2032 suggest continued urban expansion (+900 km$^2$, +5.3%), further forest loss (−376 km$^2$, −0.3%), and reductions in bareland (−27.5%) and permanent ice/snow (−19.1%).

These findings suggest that although agricultural abandonment is occurring, its signal is less visible at the national scale than the expansion of urban areas. The comparison of datasets highlights the usefulness of high-resolution products for detecting fine-scale processes and the limitations of coarser-resolution data for capturing complex land cover

transitions. The results provide information relevant to land management and policy, particularly in relation to balancing development pressures with the conservation of agricultural and forest ecosystems.

**Keywords:** Land Use Change, Remote Sensing, GRASS GIS, Urbanization, Markov Chain, Validation, Italy, ESA CCI, GLC_FCS30D, Transition Matrix.

# Abstract in lingua italiana

L'abbandono dei terreni e le successive transizioni ecologiche sono processi chiave che influenzano il paesaggio italiano, guidati da fattori socio-economici e ambientali. Questa ricerca fornisce una valutazione a scala nazionale di tali dinamiche dal 1985 al 2032, utilizzando due dataset globali di copertura del suolo: ESA CCI Land Cover (risoluzione 300 m, 1992–2022) e GLC_FCS30D (risoluzione 30 m, 1985–2022). L'analisi è stata condotta in ambiente GRASS GIS, con l'ausilio di script automatizzati per garantire coerenza e riproducibilità.

La metodologia è stata articolata in quattro componenti: (1) analisi statistica pixel-based per quantificare l'estensione annuale di ciascuna classe di copertura, con calcolo dinamico dell'area per garantire l'accuratezza cartometrica; (2) costruzione di matrici di transizione per identificare i percorsi di conversione, in particolare dalle superfici agricole a quelle naturali; (3) validazione comparativa dei due dataset, ricampionando i dati a 30 m su griglia a 300 m per valutarne la coerenza tramite matrici di confusione e metriche di accuratezza, con un accordo complessivo moderato (78–79%), risultato scarso per classi eterogenee come gli arbusteti (User's Accuracy ≈25 %), e maggiore coerenza per classi stabili come Acqua (>88 %) e Seminativi (>88 %); e (4) sviluppo di un modello di Catena di Markov vincolata, integrato con specifici fattori ecologici e socio-economici italiani, per proiettare la copertura del suolo al 2032.

I risultati indicano un paesaggio caratterizzato da elementi di persistenza e trasformazione. Nel 2022 i seminativi rappresentavano circa il 48 % del territorio nazionale e le foreste il 36 %. Le aree urbanizzate sono aumentate del 70–104 % a seconda del dataset, costituendo la trasformazione più rilevante nel periodo 1985–2022. La copertura forestale ha mostrato una lieve riduzione netta (−2.6 %), mentre le praterie hanno registrato un calo più consistente (−15.2 %). Le proiezioni al 2032 suggeriscono un'ulteriore espansione urbana (+900 km$^2$, +5.3 %), una continua riduzione delle superfici forestali (−376 km$^2$, −0.3 %) e diminuzioni delle aree a suolo nudo (−27.5 %) e dei ghiacciai/nevi permanenti (−19.1 %).

Questi risultati evidenziano che, sebbene l'abbandono agricolo sia presente, il suo seg-

nale è meno rilevante a scala nazionale rispetto all'espansione urbana. Il confronto tra dataset mette in luce l'utilità dei prodotti ad alta risoluzione per individuare processi di piccola scala e le limitazioni dei dati a risoluzione più grossolana nel descrivere transizioni complesse di copertura del suolo. I risultati forniscono elementi utili per la gestione del territorio e per le politiche di pianificazione, in particolare rispetto alla necessità di bilanciare le pressioni dello sviluppo con la conservazione dei sistemi agricoli e forestali.

**Parole chiave:** Cambiamenti di Uso del Suolo, Telerilevamento, GRASS GIS, Urbanizzazione, Catena di Markov, Validazione, Italia, ESA CCI, GLC_FCS30D, Matrice di Transizione.

# Introduction

## 1.1. Background and Motivation

Global environmental change is increasingly driven by human activities, particularly through land use and land cover (LULC) change, which is a key driver of ecosystem transformations [1]. Monitoring these changes is critical for addressing interconnected challenges related to climate change, biodiversity conservation, and sustainable resource management. Satellite-derived land cover datasets have emerged as indispensable tools for quantifying these dynamics across spatial and temporal scales, providing a synoptic view of landscape evolution [2].

Italy, situated in the heart of the Mediterranean, represents a typical case study of complex socio-ecological transitions. The country's landscape a rich mosaic shaped by millennia of human presence is undergoing a profound transformation driven largely by urbanization and agricultural land abandonment [3]. This process, fueled by rural depopulation, economic shifts, and the declining viability of traditional farming in marginal areas, initiates ecological succession from cultivated lands to shrublands and secondary forests. These spontaneous transitions carry significant and ambivalent implications: while they can enhance ecosystem services like carbon sequestration and habitat provision, they also increase landscape homogeneity, elevate wildfire risk [4], lead to decline in agricultural biodiversity, and cause loss of culturally valuable traditional landscapes [5].

Accurately mapping and understanding the rates, patterns, and trajectories of these changes is essential for informed land-use policy and effective conservation strategies. However, the remote sensing community offers a large number of global land cover (GLC) products that vary significantly in spatial resolution, thematic detail, and temporal coverage, leading to potential inconsistencies in change assessments [6]. This thesis addresses these challenges by leveraging a multi-temporal analytical framework to provide a comprehensive analysis of land cover dynamics in Italy over the past three decades.

## 1.2. Research Problem and Objectives

This thesis examines land abandonment and land cover transitions in Italy between 1985 and 2022. The complex interplay of urbanization, agricultural decline, and ecological succession in the Italian landscape necessitates a robust, multi-faceted analytical approach. To this end, this study employs an integrated computational framework built in GRASS GIS and Python to ensure both reproducibility and precision, moving from empirical observation to modelling and validation.

The research is guided by the following four primary objectives:

1. Quantify the spatiotemporal patterns of major land cover classes in Italy from 1985 to 2022, establishing clear baselines of change for cropland, forest, built-up areas, and shrubland.
2. Identify the main transition pathways between land cover classes and assess the dominant processes, such as cropland conversion to built-up or forest, across different regions.
3. Evaluate the consistency and reliability of the different GLC datasets, with particular attention to systematic misclassifications affecting heterogeneous land cover types.
4. Perform land cover evolution simulation, tailored to Italy-specific conditions, to project land cover patterns to 2032 and generate evidence-based insights for land management and planning.

To achieve these objectives, the analysis implements a sequence of complementary approaches. A pixel-based statistical analysis of annual land cover extent is conducted using both coarse- and finer-resolution datasets, employing dynamic pixel-area calculations for enhanced cartometric accuracy.

To move beyond quantification and address the second objective, transition matrix analysis is applied to trace the dominant pathways of change and unravel complex processes such as regrowth on abandoned land and forest degradation.

The third objective, concerning dataset validation, is pursued by resampling the high-resolution data to assess the thematic reliability of the coarser product through confusion matrices and accuracy metrics.

Finally, building on the insights gained, a constrained Markov Chain model is developed, integrating Italy-specific ecological and socio-economic factors to project future land cover

and meet the fourth objective. Through this multi-layered methodology, this thesis aims to provide a detailed account of land cover change in Italy while critically evaluating the analytical tools themselves, thereby offering valuable guidance for sustainable land management policy.

## 1.3. Literature Review

Land abandonment represents a profound socio-ecological shift that has significantly shaped the Italian landscape over recent decades [3]. This process is particularly prevalent in marginal and mountainous areas where traditional agricultural practices have become economically unviable. Studies estimate that substantial portions of Italian agricultural land, particularly in the Apennines and the Alps, are susceptible to abandonment, a trend driven by complex socio-economic factors including rural depopulation, an aging farmer demographic, and the declining profitability of small-scale farming [7].

The ecological implications of land abandonment in Italy are complex and multifaceted. While abandonment often initiates natural reforestation processes that can enhance carbon sequestration and promote the recovery of certain forest-dwelling species, it also introduces significant ecological trade-offs. Plieninger et al. [5] demonstrated through meta-analysis that land abandonment typically increases overall species richness (15–30%) due to vegetation succession, but simultaneously causes a decline in farmland specialist species, thereby shifting biodiversity toward more generalist assemblages. This pattern has been observed across Italian landscapes where abandoned farmlands transition through successional stages from grassland to shrubland and eventually to secondary forest [8].

In the Italian context, these transformations carry particularly important consequences for ecosystem management. The proliferation of unmanaged vegetation following abandonment has been directly linked to altered wildfire regimes across Mediterranean regions [4]. For instance, studies in Sardinia have shown that areas with high abandonment intensity experienced up to a 35% increase in wildfire occurrences due to increased fuel loads [9]. Similarly, hydrological dynamics are significantly affected as changes in vegetation cover influence evapotranspiration rates, soil moisture retention, and watershed functions [10].

Remote sensing has emerged as an essential tool for monitoring land cover changes across Italy's diverse landscapes [11]. The country's varied topography, from Alpine regions to Mediterranean coastlines, presents unique challenges for land cover classification

that have been addressed through various satellite-derived products.

The European Space Agency Climate Change Initiative Land Cover (ESA CCI) dataset (300 m resolution, 1992–2022) has been widely used for broad-scale analyses of Italian land cover dynamics [12]. Its long temporal coverage makes it particularly valuable for identifying decadal trends in forest expansion and agricultural contraction. However, its coarse spatial resolution presents limitations in Italy's fragmented landscapes, where small agricultural plots and heterogeneous forest patches require finer-scale detection capabilities [13].

The more recent Global Land Cover Fine Classification System (GLC_FCS30) dataset (30 m resolution, 1985–2022) offers improved spatial detail that better captures Italy's complex land cover patterns [14]. With an overall classification accuracy exceeding 80%, this dataset provides enhanced capability for detecting small-scale transitions particularly relevant to the Italian context, such as the conversion of olive groves to shrubland or the gradual reforestation of abandoned terraces. Nevertheless, challenges remain in classifying certain cover types; Tsendbazar et al. [6] identified persistent confusion between grassland and shrubland categories in Mediterranean environments, which directly affects the accurate monitoring of successional processes following abandonment.

The integration of multiple datasets has proven particularly valuable for comprehensive land cover assessment in Italy. The interaction between urbanization patterns and agricultural abandonment creates complex landscape dynamics that no single dataset can fully capture [13]. While global impervious surface datasets show substantial urban expansion in northern Italian regions, this urban growth occurs simultaneously with agricultural abandonment in rural areas, creating a pattern of landscape polarization that requires sophisticated analytical approaches to disentangle [15].

Despite these technological advancements, significant challenges remain in remote sensing of Italian land cover change [6]. The country's diverse bioclimatic regions, from Alpine to Mediterranean ecosystems, create spectral confusion problems in classification algorithms. These limitations necessitate robust validation frameworks and the integration of complementary data sources for accurate land cover monitoring.

The complex nature of Italian landscape transformations requires sophisticated analytical frameworks that can capture both the patterns and processes of change [16]. Transition matrices have emerged as particularly valuable tools for quantifying land cover changes in Mediterranean environments, moving beyond simple net change statistics to reveal the specific pathways of transformation. These matrices enable researchers to identify dominant processes such as urbanization versus natural succession, providing deeper

insights into landscape dynamics that are particularly relevant to the Italian context, where both processes occur simultaneously [17].

Validation methodologies have evolved significantly to address the particular challenges of Italian landscapes [18]. Inter-dataset comparisons have become increasingly important for assessing product reliability in regions with complex land cover patterns. Techniques such as confusion matrices and spatial consistency indices help identify systematic classification errors, particularly relevant for distinguishing between spectrally similar classes such as olive groves, shrubland, and young forest stands that characterize much of Italy's abandoned agricultural lands [19, 20].

More recently, predictive modeling approaches like Markov Chain analysis have been employed to project future land cover trends in Italian landscapes based on observed transition probabilities [21]. When constrained by realistic ecological and socio-economic factors specific to the Italian context such as topographic limitations, conservation policies, and regional development patterns these models can provide valuable insights into potential future scenarios under current trajectories. However, these models must account for Italy's diverse regional contexts, from the intensive agriculture of the Po Valley to the pastoral systems of the Apennines, each exhibiting distinct abandonment and transformation patterns [19].

## 1.4. Thesis Structure

The remainder of the thesis is organised into three chapters, which logically progress from defining the research problem to presenting the results and conclusions.

**Chapter 2: Materials and Methods** details the entire analytical framework. It first describes the two primary data sources, the ESA CCI LC (300 m) and GLC_FCS30D (30 m) datasets, and introduces Italy as a complex and representative Mediterranean study area. The core of the chapter is the detailed presentation of the four-pillar analytical methodology: (1) pixel-based statistical analysis for quantifying land cover extents, (2) transition matrix computation to identify change pathways, (3) inter-dataset validation to assess thematic accuracy, and (4) a constrained Markov Chain model for projecting land cover to 2032. The chapter concludes by specifying the software tools (GRASS GIS, Python) and measures taken to ensure reproducibility and quality assurance.

**Chapter 3: Results and Discussion** presents and interprets the empirical findings. It starts by comparing the land cover composition and multi-decadal trends derived from both datasets. It then delves into the land cover transition analysis, revealing the

dominant processes of urbanization and complex agro-forestry dynamics, and identifies temporal hotspots of change. Accuracy assessment is used to quantify the agreement and discrepancies between the two datasets. Finally, the results from the Markov Chain projection of land cover to 2032 are presented. The discussion section interprets these key findings, compares them with previous studies, evaluates the methodological strengths and limitations, and elaborates on their implications for land management and policy in Italy.

**Chapter 4: Conclusion and Future Work** synthesizes the study. It provides a summary of the principal findings, affirming the prevalence of urbanization over forest gain from abandonment and the critical importance of spatial resolution in change detection. The chapter also discusses the limitations of the work, particularly regarding classification uncertainty and model assumptions. It concludes by proposing directions for future research, suggesting the integration of socio-economic drivers, the development of more sophisticated hybrid models, and the need for high-resolution validation campaigns.

# Materials and Methods

Understanding long-term land cover dynamics in Italy requires reliable datasets that provide consistent and spatially explicit classifications. This chapter outlines the key datasets and methodologies used in this study to analyze land abandonment and cover transitions across Italy from 1985 to 2032 [2, 17].

## 1.5. Data Sources

The research is underpinned by two complementary GLC datasets, selected for their compatible spatial, temporal and semantic characteristics [18]:

- **ESA Climate Change Initiative (ESA CCI) Land Cover Dataset** (300m resolution, 1992–2022) [12]
- **Global Land Cover Fine Classification System (GLC_FCS30D)** (30m resolution, 1985–2022) [14]

The complementary use of both datasets, compared in Table 1, allows for a multi-scale analysis of land cover changes in Italy, leveraging the long-term consistency of ESA CCI LC and the fine spatial detail of GLC_FCS30D [6]. This approach enables detection of both broad-scale trends and fine-grained spatial patterns in land cover dynamics across the Italian landscape.

### 1.5.1. ESA Climate Change Initiative Land Cover (ESA CCI LC)

The ESA Climate Change Initiative (CCI) Land Cover dataset provides a globally consistent time series of annual land cover maps at a spatial resolution of 300 meters, spanning from 1992 to 2022 [12]. Developed by the European Space Agency, it integrates data from multiple satellite sensorsAVHRR (1992–1999), SPOT-VGT (1998–2012), MERIS (2003–2012), and PROBA-V and Sentinel-3 (from 2013 onward)to create a harmonized and temporally consistent time series.

**Table 1: Comparison of land cover datasets used in this study.**

| Characteristic | ESA CCI LC | GLC_FCS30D |
|---|---|---|
| Spatial resolution | 300 m | 30 m |
| Temporal coverage | 1992–2022 | 1985–2022 |
| Number of classes | 37 | 35 |
| Data format | NetCDF (single tile) | GeoTIFF (multiple tiles) |
| Coordinate system | WGS84 | WGS84 |
| Approx. size per year | 100 MB | 1 GB |
| Total dataset size | 3.2 GB | 30 GB |
| Primary sensors | AVHRR, SPOT-VGT, MERIS, PROBA-V, Sentinel-3 | Landsat |

The dataset includes 37 land cover classes following the UN-FAO Land Cover Classification System (LCCS), offering thematic detail such as the distinction between broadleaved and needleleaved, evergreen and deciduous forests. Its change detection is based on IPCC land categories with a multi-year observation method for consistency. Distributed in NetCDF format as single-tile global maps via the Copernicus Climate Data Store (C3S CDS), each annual layer is approximately 100 MB in size. This dataset enables the analysis of long-term trends in deforestation, reforestation, and agricultural abandonment at regional scales, making it ideal for capturing Italy's macro-scale land cover dynamics [12]. The classification legend is presented in Figure 1.

### 1.5.2. Global Land Cover with Fine Classification System at 30m (GLC_FCS30)

The GLC_FCS30D dataset offers high-resolution (30m) global land cover maps from 1985 to 2022, providing finer spatial detail essential for detecting localized changes in Italy's fragmented landscapes [14]. It builds upon the original GLC_FCS30 product by expanding its temporal coverage from two epochs to a near-annual time series. Developed using extensive Landsat imagery and continuous change detection algorithms within Google Earth Engine, it supports detailed analysis of land cover transitions.

The classification system comprises 35 land cover subcategories based on the UN-FAO LCCS with enhanced thematic detail (Figure 2). The data is projected in WGS84 and distributed in GeoTIFF format as multiple tiles. The high granularity of GLC_FCS30D

*Legend of the global CCI-LC maps, based on LCCS*

| Value | Label | Color |
|---|---|---|
| 0 | No Data | |
| 10 | Cropland, rainfed | |
| 11 | Herbaceous cover | |
| 12 | Tree or shrub cover | |
| 20 | Cropland, irrigated or post-flooding | |
| 30 | Mosaic cropland (>50%) / natural vegetation (tree, shrub, herbaceous cover) (<50%) | |
| 40 | Mosaic natural vegetation (tree, shrub, herbaceous cover) (>50%) / cropland (<50%) | |
| 50 | Tree cover, broadleaved, evergreen, closed to open (>15%) | |
| 60 | Tree cover, broadleaved, deciduous, closed to open (>15%) | |
| 61 | Tree cover, broadleaved, deciduous, closed (>40%) | |
| 62 | Tree cover, broadleaved, deciduous, open (15-40%) | |
| 70 | Tree cover, needleleaved, evergreen, closed to open (>15%) | |
| 71 | Tree cover, needleleaved, evergreen, closed (>40%) | |
| 72 | Tree cover, needleleaved, evergreen, open (15-40%) | |
| 80 | Tree cover, needleleaved, deciduous, closed to open (>15%) | |
| 81 | Tree cover, needleleaved, deciduous, closed (>40%) | |
| 82 | Tree cover, needleleaved, deciduous, open (15-40%) | |
| 90 | Tree cover, mixed leaf type (broadleaved and needleleaved) | |
| 100 | Mosaic tree and shrub (>50%) / herbaceous cover (<50%) | |
| 110 | Mosaic herbaceous cover (>50%) / tree and shrub (<50%) | |
| 120 | Shrubland | |
| 121 | Evergreen shrubland | |
| 122 | Deciduous shrubland | |
| 130 | Grassland | |
| 140 | Lichens and mosses | |
| 150 | Sparse vegetation (tree, shrub, herbaceous cover) (<15%) | |
| 151 | Sparse tree (<15%) | |
| 152 | Sparse shrub (<15%) | |
| 153 | Sparse herbaceous cover (<15%) | |
| 160 | Tree cover, flooded, fresh or brakish water | |
| 170 | Tree cover, flooded, saline water | |
| 180 | Shrub or herbaceous cover, flooded, fresh/saline/brakish water | |
| 190 | Urban areas | |
| 200 | Bare areas | |
| 201 | Consolidated bare areas | |
| 202 | Unconsolidated bare areas | |
| 210 | Water bodies | |
| 220 | Permanent snow and ice | |

Figure 1: Legend of the global ESA CCI-LC maps, based on the UN-FAO Land Cover Classification System (LCCS). Source: [12].

provides capability for detecting fine-scale processes like forest degradation to shrubland, small urban patches, and agricultural fragmentation [14].

■ **The fine classification system containing 35 land-cover sub-categories**

| LC id | Classification System | Color |
|---|---|---|
| 10 | Rainfed cropland | (255,255,100) |
| 11 | Herbaceous cover cropland | (255,255,100) |
| 12 | Tree or shrub cover (Orchard) cropland | (255,255,0) |
| 20 | Irrigated cropland | (170,240,240) |
| 51 | Open evergreen broadleaved forest | (76,115,0) |
| 52 | Closed evergreen broadleaved forest | (0,100,0) |
| 61 | Open deciduous broadleaved forest (0.15<fc<0.4) | (170,200,0) |
| 62 | Closed deciduous broadleaved forest (fc>0.4) | (0,160,0) |
| 71 | Open evergreen needle-leaved forest (0.15< fc <0.4) | (0,80,0) |
| 72 | Closed evergreen needle-leaved forest (fc >0.4) | (0,60,0) |
| 81 | Open deciduous needle-leaved forest (0.15< fc <0.4) | (40,100,0) |
| 82 | Closed deciduous needle-leaved forest (fc >0.4) | (40,80,0) |
| 91 | Open mixed leaf forest (broadleaved and needle-leaved) | (160,180,50) |
| 92 | Closed mixed leaf forest (broadleaved and needle-leaved) | (120,130,0) |
| 120 | Shrubland | (150,100,0) |
| 121 | Evergreen shrubland | (150,75,0) |
| 122 | Deciduous shrubland | (150,100,0) |
| 130 | Grassland | (255,180,50) |
| 140 | Lichens and mosses | (255,220,210) |
| 150 | Sparse vegetation (fc<0.15) | (255,235,175) |
| 152 | Sparse shrubland (fc<0.15) | (255,210,120) |
| 153 | Sparse herbaceous (fc<0.15) | (255,235,175) |
| 181 | Swamp | (0,168,132) |
| 182 | Marsh | (115,255,223) |
| 183 | Flooded flat | (158,187,215) |
| 184 | Saline | (130,130,130) |
| 185 | Mangrove | (245,122,182) |
| 186 | Salt marsh | (102,205,171) |
| 187 | Tidal flat | (68,79,137) |
| 190 | Impervious surfaces | (195,20,0) |
| 200 | Bare areas | (255,245,215) |
| 201 | Consolidated bare areas | (220,220,220) |
| 202 | Unconsolidated bare areas | (255,245,215) |
| 210 | Water body | (0,70,200) |
| 220 | Permanent ice and snow | (255,255,255) |
| 0, 250 | Filled value | (255,255,255) |

**Figure 2: The Fine Classification System (FCS) legend used in the GLC_FCS30D product. Source: Adapted from [14].**

## 1.6. Dataset Integration Strategy

The selected datasets were integrated to leverage their complementary strengths in spatial resolution, temporal coverage, and classification accuracy [6]. The ESA CCI dataset was utilized for its consistent global coverage to facilitate long-term trend analysis over three decades (1992–2022) [12]. Concurrently, the GLC_FCS30D dataset provided high-resolution detail, which is essential for understanding fine-grained, localized changes within Italy's heterogeneous landscapes [14]. Furthermore, this high-resolution dataset can served as a reference for validation to assess the thematic reliability of ESA CCI through resampling and confusion matrix analysis [18] thanks to its significantly higher resolution with respect to the CCI product. The GLC_FCS30D also extended the temporal scope of the analysis back to 1985, providing additional historical context for

the observed land cover dynamics [14]. This multi-resolution, multi-temporal integration support the detection of both long-term patterns and short-term dynamics across Italy's diverse ecosystems, from Alpine regions to Mediterranean coastlines [22].

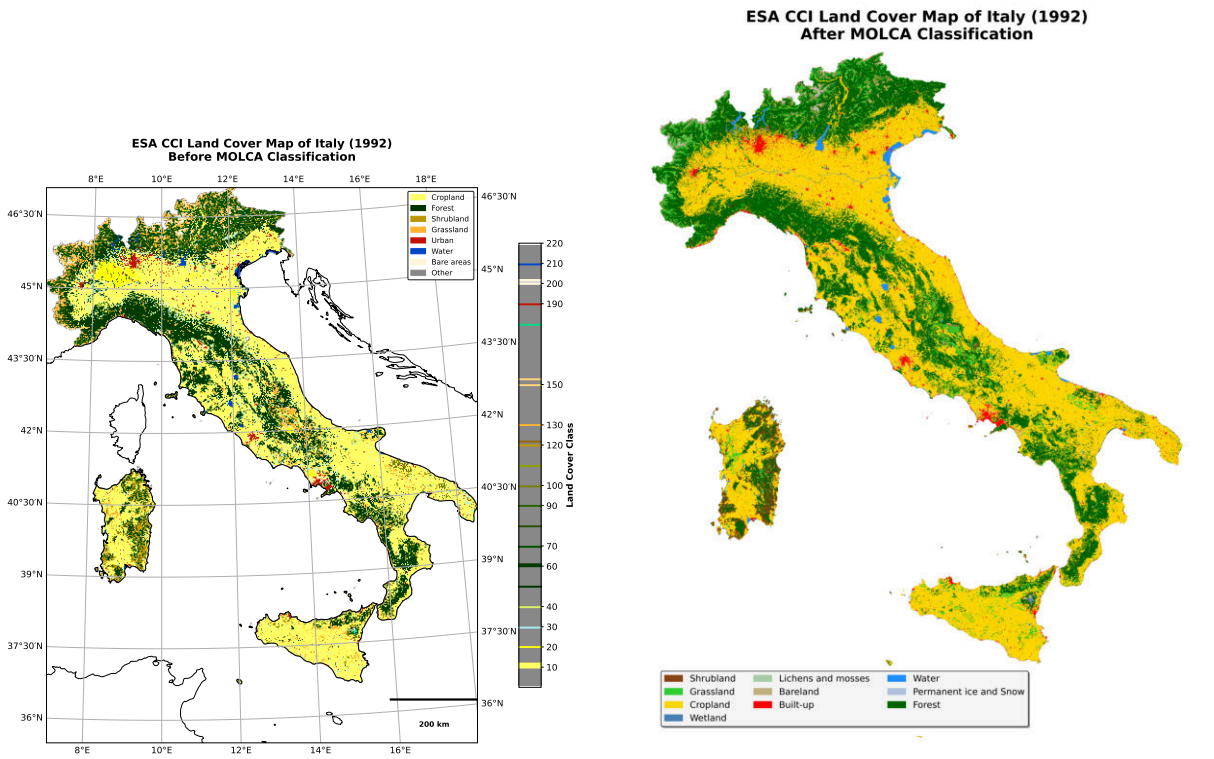
### 1.6.1. Land Cover Legend Harmonization

Land cover data from the ESA CCI and the GLC_FCS30D were collected for the time periods 1992–2022 and 1985–2022, respectively. The datasets were pre-processed using QGIS and Python. Initial steps included clipping the data to the study area and reprojecting them to a common coordinate reference system (EPSG:4326 - WGS84) to ensure spatial alignment. Co-registration and resampling techniques were applied to harmonize spatial resolution and extent across time. The land cover classes were then reclassified into MOLCA (Modified Land Cover Aggregation) classes to standardize classification schemes and simplify further analysis. The MOLCA derivation procedure was designed, developed, and successfully tested within the ESA High-Resolution Land Cover CCI project [23]. This enabled the computation of class-wise area statistics, which are summarized in Table 2 and Figure 3.

**MOLCA Classification Table with Color Legend**

| MOLCA Code | MOLCA Label | CCI LC Code | CCI LC Labels | GLC_FCS30D Code | GLC_FCS30D Labels |
|---|---|---|---|---|---|
| 20 | Forest | 50–100 | Tree cover types | 51–92 | Open/Closed forests |
| 5 | Shrubland | 120–152 | Shrubland; Evergreen/Deciduous | 120–152 | Shrubland |
| 7 | Grassland | 110; 130; 153 | Grassland cover types | 130; 153 | Grassland types |
| 8 | Cropland | 10–40 | Rainfed/Irrigated cropland | 10; 11; 12; 20 | Herbaceous/orchard cropland |
| 9 | Wetland | 160–180 | Flooded cover types | 181–187 | Swamp, Marsh, Mangrove |
| 11 | Lichens and mosses | 140 | Lichens and Mosses | 140 | Lichens and Mosses |
| 12 | Bareland | 200–202 | Bare Areas | 200–202 | Same as CCI |
| 13 | Built-up | 190 | Urban areas | 190 | Impervious surfaces |
| 15 | Water | 210 | Water bodies | 210 | Water body |
| 16 | Permanent ice and Snow | 220 | Permanent snow and ice | 220 | Permanent snow and ice |
| 255 | No data | 0 | No data | 250 | Filled value |

**Table 2: Legend harmonization using the MOLCA reclassification schema, showing the correspondence between original class codes and labels from the ESA CCI and GLC_FCS30D datasets and the unified MOLCA classes.**




**(a) Italy Land Cover Map (ESA CCI - 1992, before MOLCA classification)**

**(b) Italy Land Cover Map (ESA CCI - 1992, after MOLCA classification)**

**Figure 3: Comparison of ESA CCI Land Cover data for Italy (1992) before and after applying the MOLCA legend classification system. The MOLCA processing standardizes the legend to support dataset comparison.**

## 1.7. Study Area

Italy was considered as the study area in this work for analyzing land cover transitions due to its diverse bioclimatic gradients, which range from Alpine to semi-arid ecosystems [22, 24]. The country's complex topography and long history of human settlements have produced a dynamic landscape mosaic characterized by contrasting change trajectories. These include agricultural abandonment in mountainous areas, urban expansion along coastlines, agricultural intensification in lowlands, and widespread forest recovery [3, 4]. This environmental heterogeneity, encompassing elevations from coastal plains to the high Alps (4,810 m), provides an ideal setting to investigate how geographical factors mediate land abandonment processes [8].

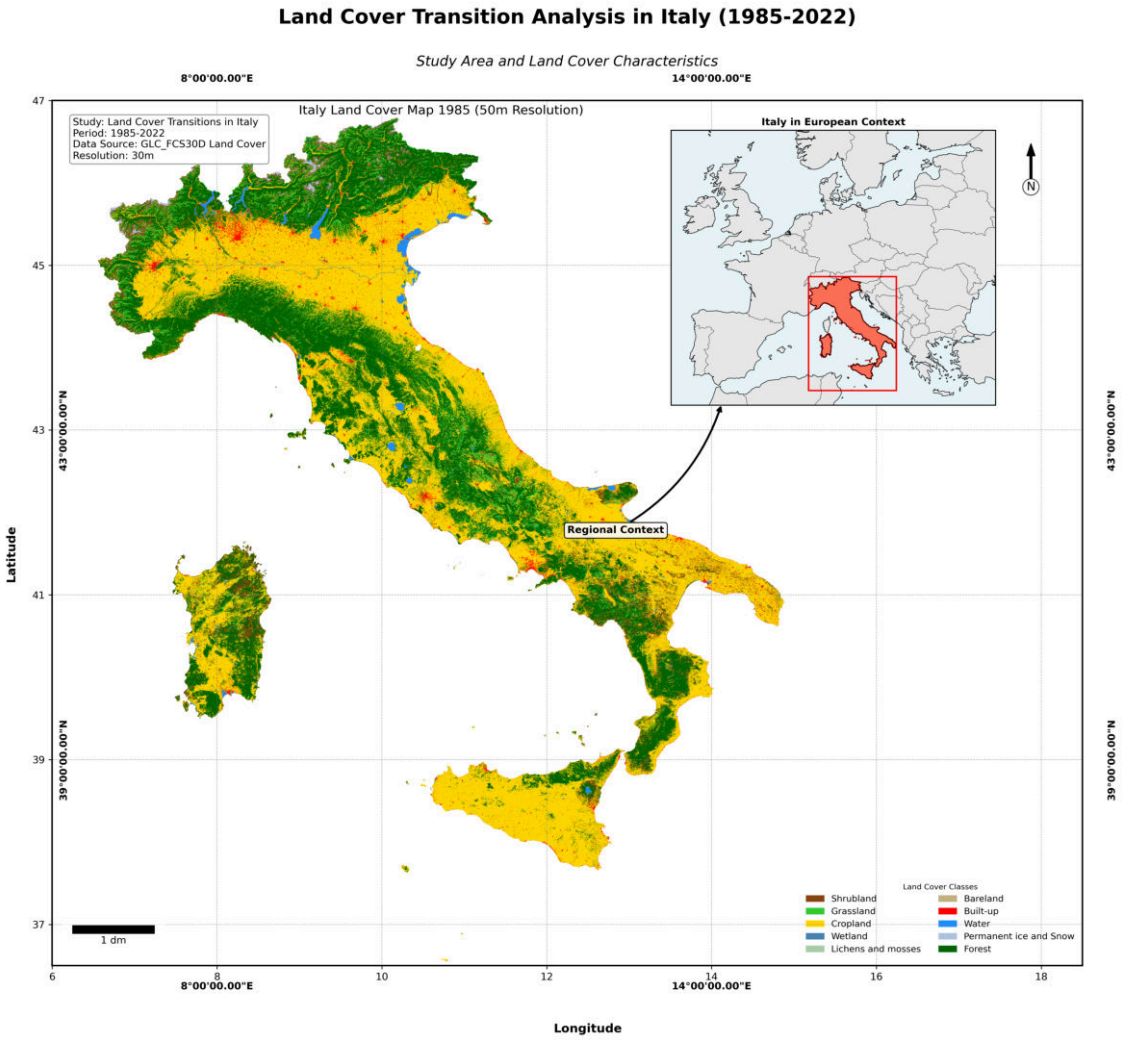



**Figure 4: Study Area: Italy showing diverse topography and land cover patterns.**

## 1.8. Data Preprocessing

The methodological approach follows a workflow designed to ensure consistency and accuracy in land cover change analysis [18]. The research implements four core analytical components as outlined below.

### 1.8.1. Data Acquisition and Preprocessing

The methodological workflow begins with data acquisition and preprocessing [18]:

1. **Data Acquisition:** Land cover datasets were obtained from the European Space Agency's Climate Change Initiative (ESA CCI 1992–2022) and the Global Land

Cover dataset with Fine Classification System at 30 meters (GLC_FCS30D, 1985–2022) [12, 14].

2. **Clipping to Study Area:** The datasets were spatially clipped to Italy's geographical extent using QGIS and Python-based geospatial libraries.
3. **Reprojection to Common CRS:** All raster layers were reprojected to EPSG:3035 (ETRS89-extended / LAEA Europe) to ensure spatial alignment and metric area calculations.
4. **Land Cover Reclassification:** Original land cover classes were reclassified into standardized categories following the MOLCA scheme for thematic harmonization.
5. **Co-registration and Resampling:** Co-registration techniques aligned spatial extents, with datasets resampled to common resolutions, used for accuracy assessment and dataset inter-comparison [18].

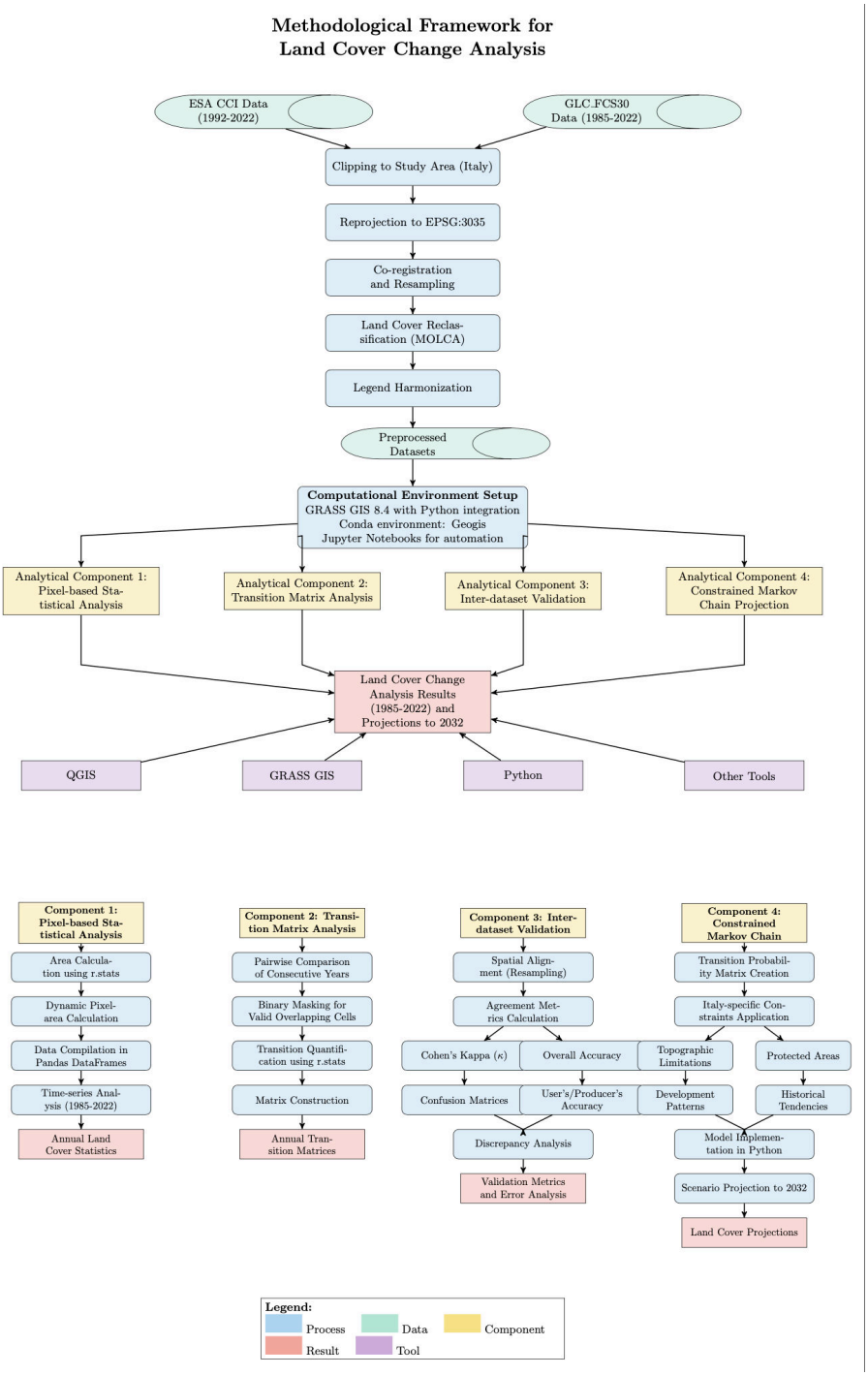



**Figure 5: Methodological Framework for land cover change analysis**

## 1.9. Analytical Framework

The analytical framework of this study integrates map algebra and cross-statistics approaches to quantify and validate land cover changes in Italy from 1985 to 2022. It

combines pixel-level statistics, transition analysis, inter-dataset comparisons and future prediction modelling using full reproducible strategies mostly based on free and open source software such as Python and GRASS GIS [25].

Annual land cover extents were quantified using pixel-based statistics [18]. Area calculations were performed with the GRASS GIS module `r.stats`, including cartometrically accurate computations to account for projection distortions. Results were structured in Pyhton `pandas` DataFrames containing total area ($km^2$), proportional share (%), and pixel counts. Annual statistics from 1985 to 2022 enabled trend identification [26].

The quantitative assessment of land cover dynamics was performed through a systematic computational procedure. The core of this procedure involved the calculation of four fundamental metrics for each land cover class $c$ in each year $t$.

The analysis begins with the extraction of the raw pixel count ($N_{c,t}$) for each land cover class $c$ from a given raster map $R_t$ using the GRASS GIS module `r.stats`. This value represents the most basic unit of analysis. To convert these pixel counts into areas, the exact pixel area ($A_{pixel}$) was dynamically calculated based on the current geographic region's north-south and east-west resolution settings. This calculation, defined as $A_{pixel} = nsres \times ewres$, was critical for maintaining metric accuracy within the projected coordinate system (ETRS89-LAEA). The areal extent in square meters for each class was then derived by multiplying the pixel count by this pixel area:

$$A_{c,t}^{m^2} = N_{c,t} \times A_{pixel}. \tag{1}$$

For easier interpretation and reporting at a regional scale, the areal extent was converted from square meters to square kilometers:

$$A_{c,t}^{km^2} = \frac{A_{c,t}^{m^2}}{10^6}. \tag{2}$$

Finally, to enable a normalized comparison of class dominance over time, the proportional coverage ($P_{c,t}$) of each class was computed. This metric expresses the area of a class as a percentage relative to the total study area ($T_t$) for that year:

$$P_{c,t} = \left( \frac{A_{c,t}^{km^2}}{T_t} \right) \times 100\% \quad \text{where} \quad T_t = \sum_{c=1}^{n} A_{c,t}^{km^2}. \tag{3}$$

This procedure was executed iteratively for every raster in the temporal series. The

results were structured into a data matrix, which served as the primary input for all subsequent change detection algorithms and trend analyses, forming a summary dataset for assessing land cover dynamics.

Annual transitions between land cover classes were decoded using per-pixel change detection [27]. Consecutive yearly rasters were compared, and valid overlapping pixels isolated through raster algebra. Transition computations using GRASS GIS `r.stats` were converted to area ($km^2$) and aggregated into matrices identifying dominant processes.

Transition matrices were used to quantify the spatial changes between consecutive time periods, moving beyond net change to reveal the specific pathways of conversion. This process was automated to systematically compare each annual map with its subsequent one in the temporal series.

The purpose of this analysis was the creation of a transition map for each consecutive pair of years ($t$ and $t+1$). This was achieved by algorithmically combining the two classified rasters ($R_t$ and $R_{t+1}$) into a single composite layer where each pixel's value uniquely encodes its land cover history. The composite value was generated by scaling the class code from the initial year and adding the code from the final year:

$$T_{t \to t+1} = C_t \times 1000 + C_{t+1}, \tag{4}$$

where $C_t$ and $C_{t+1}$ are the class codes for a pixel at time $t$ and $t+1$, respectively. This encoding ensures that a unique identifier represents every possible transition, such as Forest (20) to Built-up (13) being stored as the numerical value 2013.

Subsequently, the areal extent of each unique transition category was computed. The area of a transition ($A^{m^2}_{t \to t+1}$) from class $i$ to class $j$ was calculated by summing the areas of all pixels sharing the same encoded value, with the area of an individual pixel ($A_{pixel}$) derived from the raster resolution:

$$A^{m^2}_{i \to j} = N_{i \to j} \times A_{pixel} \quad \text{where} \quad A_{pixel} = nsres \times ewres. \tag{5}$$

This area was then standardized to square kilometers for reporting ($A^{km^2}_{i \to j} = A^{m^2}_{i \to j}/10^6$). The result for each time interval was a transition matrix, a two-dimensional contingency table where rows represent the land cover classes at time $t$ and columns represent the classes at time $t+1$. The cells of this matrix contain the values $A^{km^2}_{i \to j}$, providing a complete account of all gains, losses, and persistence.

This procedure was executed iteratively for each consecutive pair of years in the time

series (e.g., 1985–1990, 1990–1995, . . . , 2021–2022). The output for each interval included a CSV file listing every observed transition with its corresponding area and a GeoTIFF raster map visually representing the spatial distribution of these changes across the landscape. The full set of annual transition matrices forms the base data product enabling analysis of transition probabilities, dominant change pathways, and the dynamics of land cover conversion.

Inter-comparison between ESA CCI and GLC_FCS30D datasets was performed through standard cross-tabular statistics derived from classification confusion matrices [6, 18]. GLC_FCS30D rasters were resampled to match ESA CCI resolution using nearest-neighbor interpolation. Agreement metrics included Cohen's Kappa ($\kappa$), overall accuracy, confusion matrices, and user's/producer's accuracy. Discrepancy analysis identified systematic errors and frequently confused classes [6].

The methodological approach first required bringing co-registration of both datasets. The high-resolution maps ($R_{high}$) were resampled to the 300m grid of the reference dataset ($R_{low}$) using a majority filter (mode), which assigned the most frequent land cover class within each coarse pixel. This resampled version ($R_{resampled}$) facilitated a direct, pixel-to-pixel comparison. A binary agreement map was then generated to visually demarcate areas of concordance and discordance:

$$A = \begin{cases} 1 & \text{if } R_{low} = R_{resampled} \\ 0 & \text{if } R_{low} \neq R_{resampled} \end{cases}, \tag{6}$$

where a value of 1 (agreement) indicates that the reference and the resampled classification assigned the same label to a pixel.

The core of the quantitative assessment was the generation of an error matrix (or confusion matrix) $M$, where the rows represented the class labels from the reference dataset ($R_{low}$) and the columns represented the labels from the resampled high-resolution map ($R_{resampled}$). Each cell $m_{ij}$ in this matrix contained the number of pixels classified as class $i$ in the reference data and class $j$ in the resampled data.

From this error matrix, three key accuracy metrics were derived for each year analysed. The Overall Accuracy ($OA$) quantified the proportion of correctly classified pixels across all classes:

$$OA = \frac{\sum_{k=1}^{n} m_{kk}}{N}, \tag{7}$$

where $n$ is the number of classes, $m_{kk}$ are the diagonal elements of the matrix (correct classifications), and $N$ is the total number of pixels. The Kappa coefficient ($\kappa$) was

calculated to measure the agreement between the two datasets:

$$\kappa = \frac{OA - p_e}{1 - p_e}. \tag{8}$$

Here, $p_e$ is the hypothetical probability of chance agreement. Furthermore, class-specific accuracies were computed. The Producer's Accuracy ($PA_i$), or recall, measured the probability that a reference pixel of class $i$ was correctly classified:

$$PA_i = \frac{m_{ii}}{m_{i+}}, \tag{9}$$

where $m_{i+}$ is the total number of pixels of class $i$ in the reference data (the row total). Conversely, the User's Accuracy ($UA_j$), or precision, measured the probability that a pixel classified as class $j$ in the resampled data actually represents that class on the ground:

$$UA_j = \frac{m_{jj}}{m_{+j}}, \tag{10}$$

where $m_{+j}$ is the total number of pixels classified as class $j$ in the resampled data (the column total).

This comprehensive accuracy assessment was performed for multiple years, generating a time series of consistency metrics that evaluated not only the classification's absolute accuracy but also its reliability and temporal stability compared to a benchmark dataset.

Building upon the comprehensive accuracy assessment of land cover datasets, this part presents a predictive analysis of land cover changes in Italy from 2022 to 2032 using an enhanced Markov Chain approach integrated with realistic environmental constraints. This methodology leverages historical change patterns identified in previous analyses to generate spatially explicit and physically plausible projections of Italy's evolving landscape.

To project future land cover dynamics, a constrained Markov Chain model was developed [28, 29]. This approach integrates empirically derived transition probabilities with domain-specific knowledge to generate realistic and spatially explicit forecasts [30]. The projection was built upon the historical transition matrix $P$, calculated from observed changes in the GLC_FCS30D dataset between 2010 and 2022, which revealed a total change area of 19,536 km$^2$ (6.5% of Italy's land area) [31].

The future state for a target year was calculated by projecting the current state vector

$S_t$ through the transition matrix raised to the power of the number of time steps $k$:

$$S_{t+k} = S_t \times P^k. \tag{11}$$

To ensure geographical plausibility, this statistical projection was refined through the application of physio-geographical and socio-economic constraints tailored to the Italian context [32, 33]. This constrained the projected area ($A_i^{predicted}$) of each class $i$ within realistic maximum ($C_{max,i}$) and minimum ($C_{min,i}$) limits derived from national statistics and expert knowledge [34]:

$$A_i^{predicted} \leq C_{max,i} \tag{12}$$

$$A_i^{predicted} \geq C_{min,i}. \tag{13}$$

Furthermore, the model incorporated realistic annual change rates ($r_i$) for key classes from FAO data, blending the Markov-derived prediction with a trend-based forecast [35]. The final prediction enforced the conservation of total area ($A_{total}$) to match the known area of Italy:

$$A_i^{final} = A_i^{predicted} \times \frac{A_{total}}{\sum_i A_i^{predicted}}. \tag{14}$$

The output of this process was a spatially explicit projection map for 2032, created by applying the most probable transitions to the 2022 land cover map via spatial reclassification in GRASS GIS. Quality assurance and reproducibility were maintained through automated Python/GRASS workflows and comprehensive validation protocols [6]. This provided not only a quantitative forecast of changes but also a visual representation of their probable locations, offering a valuable tool for strategic environmental planning.

```
conda activate Geogis
export GISBASE="/Applications/GRASS-8.4.app/Contents/Resources"
export PATH="$GISBASE/bin:$GISBASE/scripts:$PATH"
export PYTHONPATH="$GISBASE/etc/python:$PYTHONPATH"
export GISDBASE="$HOME/grassdata"
export LOCATION_NAME="italy_study"
export MAPSET="landcover_analysis"
```

As part of the analysis, we used code deposited in the repository by Zafari [36] to process landcover dynamic transitions in Italy using global land cover datasets.

# Results

## 1.10. Land Cover Composition and Trends in Italy

This section presents the findings from a multi-temporal and multi-scale land cover change analysis for Italy, spanning the period from 1985 to 2022, by using both the ESA CCI and the GLC_FCS30D datasets. The results detail Italy's fundamental landscape composition, its evolution over nearly four decades, and the major transition patterns observed, supported by comprehensive visual analyses.

A comparative assessment of both datasets for the year 2022 reveals Italy's core landscape composition while also highlighting systematic differences in their estimations, as detailed in Table 3. Cropland emerges as the consistently dominant land cover type across the national territory, though its estimated extent varies significantly between the two products. The ESA CCI LC dataset reported a higher cropland coverage of 53.25%, compared to the 48.08% estimated by the GLC_FCS30D. Conversely, forest coverage was estimated to be higher by GLC_FCS30D (35.94%) than by ESA CCI LC (29.44%). These discrepancies are attributable to several methodological factors, including spatial resolution effects (where ESA CCI's 300m resolution may overestimate homogeneous classes through mixed pixel effects, while GLC_FCS30D's 30m resolution captures finer spatial detail), differences in classification schemes and class definitions, variations in temporal compositing periods, and distinct algorithmic approaches used in the production of each dataset [14, 37]. Despite these absolute differences in coverage estimation for specific classes, both datasets consistently captured the direction and relative magnitude of the key land cover trends over their respective periods, providing a robust, multi-source foundation for analyzing landscape dynamics.

The most consistent change observed across both datasets was the substantial and widespread expansion of built-up areas. As quantified in Tables 4 and 5, built-up areas more than doubled according to both sources, representing the most transformative process in the Italian landscape. The ESA CCI LC data recorded an increase from 2.22% (6,704.82 $km^2$) of the national territory in 1992 to 4.56% (13,738.77 $km^2$) in 2022, consti-

tuting a net gain of 7,034 km$^2$ or a relative increase of +104.91%. The GLC_FCS30D data, which covers a longer time frame, showed an expansion from 3.34% (10,065.20 km$^2$) in 1985 to 5.68% (17,120.87 km$^2$) in 2022, a net gain of 7,056 km$^2$ (+70.09%).

**Table 3: Comparative land cover percentages from ESA CCI LC and GLC_FCS30D datasets for 2022.**

| Land Cover Class | ESA CCI LC | GLC_FCS30D | Absolute Difference | Relative Difference (%) |
|---|---|---|---|---|
| Cropland | 53.25% | 48.08% | +5.17% | +9.71 |
| Forest | 29.44% | 35.94% | -6.50% | -22.08 |
| Grassland | 6.43% | 4.66% | +1.77% | +27.53 |
| Shrubland | 2.79% | 3.70% | -0.91% | -32.62 |
| Built-up | 4.56% | 5.68% | -1.12% | -24.56 |
| Bareland | 2.06% | 0.21% | +1.85% | +89.81 |
| Water | 1.24% | 1.08% | +0.16% | +12.90 |

**Table 4: Land cover trends from ESA CCI LC analysis (1992–2022).**

| Land Cover Class | 1992 Area (km$^2$) | 2022 Area (km$^2$) | Net Change (km$^2$) | Relative Change (%) |
|---|---|---|---|---|
| Cropland | 163,372.77 | 160,500.96 | -2,871.81 | -1.76 |
| Forest | 91,712.88 | 88,732.98 | -2,979.90 | -3.25 |
| Grassland | 18,257.31 | 19,377.90 | +1,120.59 | +6.14 |
| Shrubland | 7,752.24 | 8,408.16 | +655.92 | +8.46 |
| Built-up | 6,704.82 | 13,738.77 | +7,033.95 | +104.91 |
| Bareland | 9,007.38 | 6,201.54 | -2,805.84 | -31.15 |
| Water | 3,888.00 | 3,732.39 | -155.61 | -4.00 |
| Wetland | 351.81 | 354.51 | +2.70 | +0.77 |
| Permanent Ice/Snow | 345.96 | 345.96 | 0.00 | 0.00 |

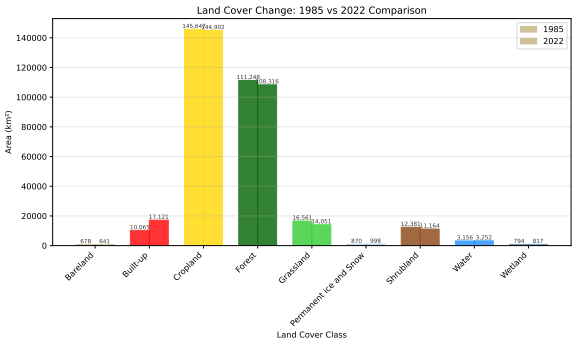
**Table 5: Land cover trends from GLC_FCS30D analysis (1985–2022).**

| Land Cover Class | 1985 Area (km²) | 2022 Area (km²) | Net Change (km²) | Relative Change (%) |
|---|---|---|---|---|
| Cropland | 145,646.94 | 144,901.97 | -744.97 | -0.51 |
| Forest | 111,247.85 | 108,316.17 | -2,931.68 | -2.64 |
| Grassland | 16,561.07 | 14,050.97 | -2,510.10 | -15.16 |
| Shrubland | 12,380.71 | 11,163.78 | -1,216.93 | -9.83 |
| Built-up | 10,065.20 | 17,120.87 | +7,055.67 | +70.09 |
| Water | 3,155.91 | 3,252.38 | +96.47 | +3.06 |
| Bareland | 678.10 | 641.41 | -36.69 | -5.41 |
| Wetland | 793.52 | 816.81 | +23.29 | +2.94 |
| Permanent Ice/Snow | 870.24 | 997.82 | +127.58 | +14.66 |
| Lichens/Mosses | 0.00 | 133.88 | +133.88 | +100.00 |

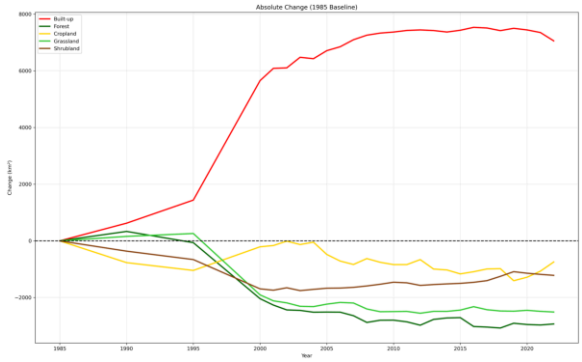
Both datasets also agree on a gradual yet persistent decline in both agricultural and forest areas, though the magnitude of loss differs. Cropland showed a gradual decline in both datasets: ESA CCI LC recorded a decrease from 54.21% to 53.25% (-1.76% change), while GLC_FCS30D showed a more marginal decrease from 48.32% to 48.08% (-0.51%). Similarly, forest cover exhibited a moderate decrease: from 30.43% to 29.44% (-3.25%) according to ESA CCI LC, and from 36.91% to 35.94% (-2.64%) according to GLC_FCS30D.

The most notable inter-dataset discrepancy concerns the trends in grassland and shrubland. The ESA CCI LC data indicates moderate increases in both grassland (+6.14%) and shrubland (+8.46%), a pattern that suggests potential processes of agricultural abandonment and subsequent natural revegetation or successional growth [27]. In direct contrast, the GLC_FCS30D data shows significant declines in both grassland (-15.16%) and shrubland (-9.83%). This fundamental divergence underscores the well-documented challenges in consistently classifying transitional, heterogeneous, and often mixed vegetation classes across different spatial resolutions, classification systems, and class definitions [18].

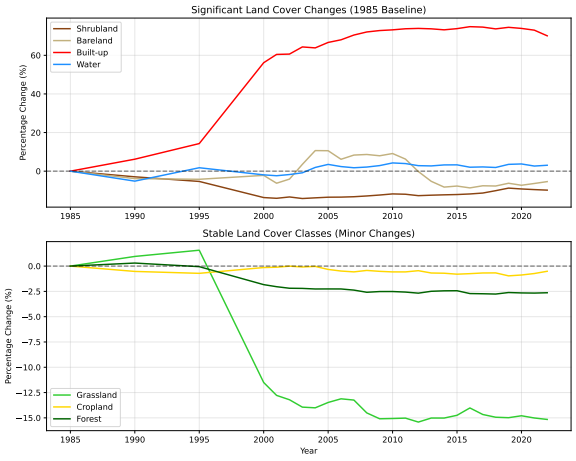
The overarching land cover trends are visualized comprehensively in Figures 6 and 7, which provide a summary overview of distribution comparisons, absolute and relative changes, and temporal trends. The visual analysis confirms the massive scale of urbanization, visible as a dominant signal in the change maps, often at the expense of agricultural land on the peri-urban fringe. The relative stability of the core cropland areas, despite the net national decrease, is also apparent.



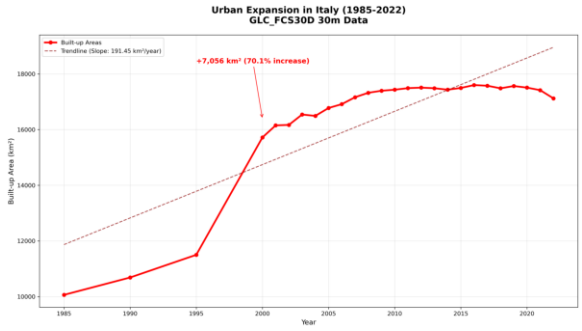

(a) Land cover distribution comparison (1985 vs. 2022)




(b) Absolute change in land cover classes (kmš)

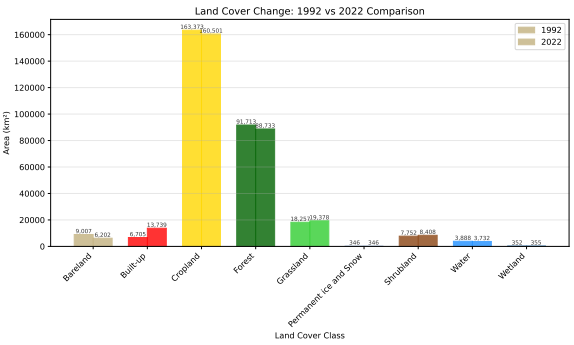



(c) Percentage change in land cover classes

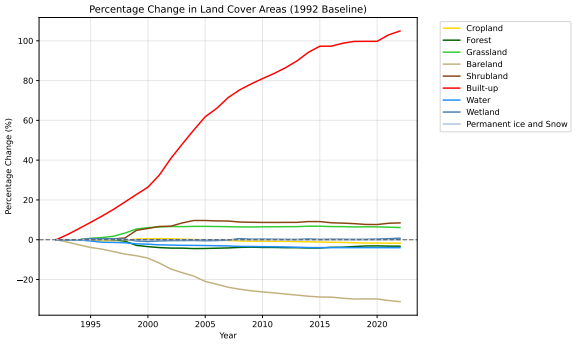



(d) Urbanization trend analysis

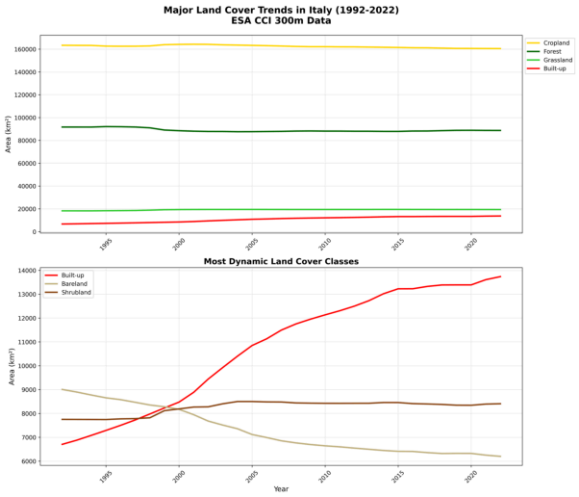
**Figure 6: GLC_FCS30D dataset analysis showing (a) 1985-2022 distribution comparison, (b) absolute area changes, (c) percentage changes, and (d) urbanization trends.**




**(a) Land cover distribution comparison (1992 vs. 2022)**




**(b) Percentage change in land cover classes**

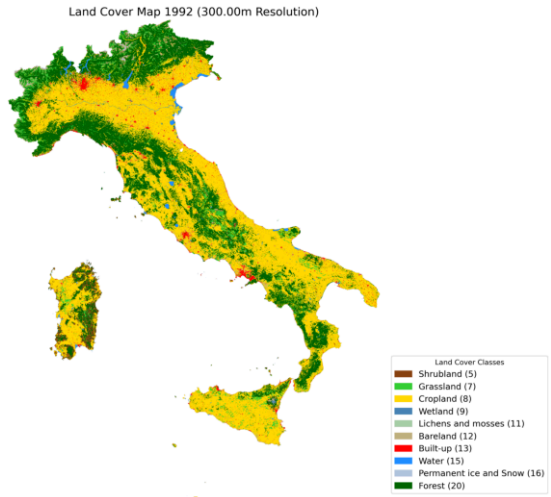



**(c) Land cover trends over time**

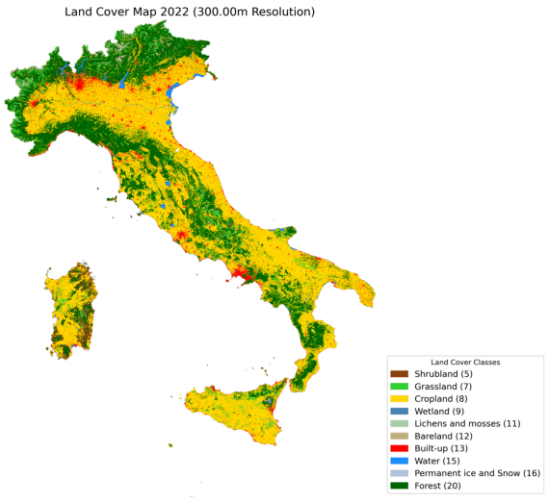
**Figure 7: ESA CCI LC dataset analysis showing (a) 1992-2022 distribution comparison, (b) percentage changes, and (c) temporal trends.**

The land cover distribution derived from the ESA CCI LC dataset is visualized in Figure 8, which shows the 1992 and 2022 distributions at 300m resolution. The 1992 map reveals the broad-scale distribution of major land cover classes at the beginning of the time series. A more detailed breakdown of this dataset is provided in Figure 9. The land cover statistics for 1992 and 2022 (Figure 10) highlight both the stability and the subtle shifts in Italy's landscape over three decades [27]. In 1992, croplands dominated with 54.2% of the national territory, followed by forests (30.4%) and grasslands (6.1%). By 2022, croplands (53.3%) and forests (29.4%) had slightly decreased, while grasslands (6.4%) and built-up areas had expanded markedly, with the latter nearly doubling from 2.2% to 4.6%. These results were computed directly from the ESA CCI LC dataset using `GRASS GIS` (`r.stats` with per-pixel area calculation) and processed via a custom `Python` workflow (using `pandas` and `tabulate`) to derive precise pixel areas, convert to km$^2$, and

calculate relative percentages [18].

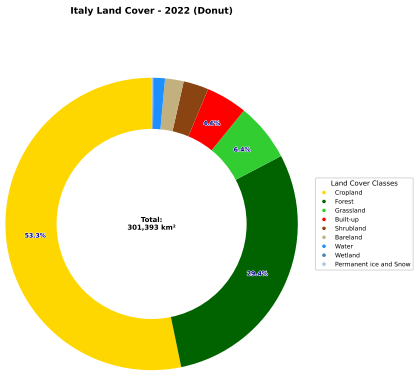



(a) Italy Land Cover Map 1992 (300m resolution)

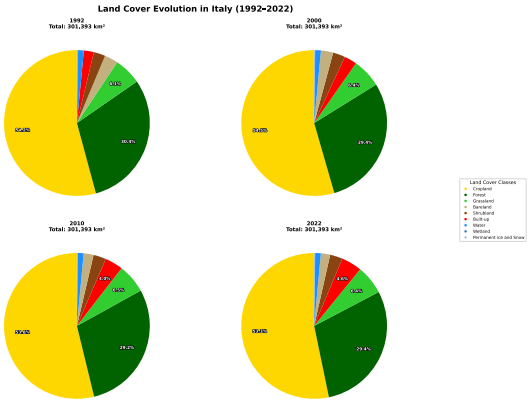



(b) Italy Land Cover Map 2022 (300m resolution)

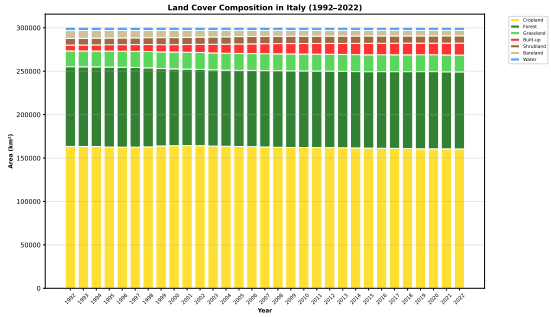
**Figure 8: ESA CCI LC land cover maps of Italy showing (a) 1992 and (b) 2022 distributions at 300m resolution.**



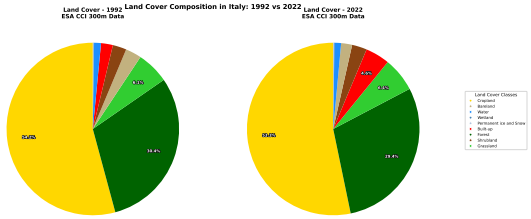

(a) Land cover distribution in Italy (2022)




(b) Multi-year land cover trends




(c) Land cover composition by stacked bar chart




(d) Comparative land cover composition (1992 vs. 2022)

**Figure 9: ESA CCI land cover analysis for Italy showing (a) 2022 distribution, (b) multi-year trends, (c) temporal composition, and (d) 1992-2022 comparison.**

Land Cover Statistics 1992

| | Year | Class | Label | Count | Area_km2 | Area_m2 | Percentage |
|---|---|---|---|---|---|---|---|
| 0 | 1992 | 8 | Cropland | 1815253.00 | 163372.77 | 163372770000.00 | 54.21 |
| 1 | 1992 | 20 | Forest | 1019032.00 | 91712.88 | 91712880000.00 | 30.43 |
| 2 | 1992 | 7 | Grassland | 202859.00 | 18257.31 | 18257310000.00 | 6.06 |
| 3 | 1992 | 12 | Bareland | 100082.00 | 9007.38 | 9007380000.00 | 2.99 |
| 4 | 1992 | 5 | Shrubland | 86136.00 | 7752.24 | 7752240000.00 | 2.57 |
| 5 | 1992 | 13 | Built-up | 74498.00 | 6704.82 | 6704820000.00 | 2.22 |
| 6 | 1992 | 15 | Water | 43200.00 | 3888.00 | 3888000000.00 | 1.29 |
| 7 | 1992 | 9 | Wetland | 3909.00 | 351.81 | 351810000.00 | 0.12 |
| 8 | 1992 | 16 | Permanent ice and Snow | 3844.00 | 345.96 | 345960000.00 | 0.11 |

(a) Land Cover Statistics (1992)

Land Cover Statistics 2022

| | Year | Class | Label | Count | Area_km2 | Area_m2 | Percentage |
|---|---|---|---|---|---|---|---|
| 0 | 2022 | 8 | Cropland | 1783344.00 | 160500.96 | 160500960000.00 | 53.25 |
| 1 | 2022 | 20 | Forest | 985922.00 | 88732.98 | 88732980000.00 | 29.44 |
| 2 | 2022 | 7 | Grassland | 215310.00 | 19377.90 | 19377900000.00 | 6.43 |
| 3 | 2022 | 13 | Built-up | 152653.00 | 13738.77 | 13738770000.00 | 4.56 |
| 4 | 2022 | 5 | Shrubland | 93424.00 | 8408.16 | 8408160000.00 | 2.79 |
| 5 | 2022 | 12 | Bareland | 68906.00 | 6201.54 | 6201540000.00 | 2.06 |
| 6 | 2022 | 15 | Water | 41471.00 | 3732.39 | 3732390000.00 | 1.24 |
| 7 | 2022 | 9 | Wetland | 3939.00 | 354.51 | 354510000.00 | 0.12 |
| 8 | 2022 | 16 | Permanent ice and Snow | 3844.00 | 345.96 | 345960000.00 | 0.11 |

(b) Land Cover Statistics (2022)

**Figure 10: Comparative land cover statistics from the ESA CCI dataset showing (a) 1992 and (b) 2022 distributions for temporal analysis.**

Complementing this broad-scale view, the GLC_FCS30D dataset provides detailed land cover maps at 30-meter resolution (Figure 13), enabling a precise visualization of

Italy's landscape patterns and fine-scale changes. The 1985 land cover map shows Italy's landscape at the beginning of the study period, characterized by extensive agricultural areas, particularly in the Po Valley and coastal plains. By 2022, the transformations are evident: urban expansion is clearly visible around major metropolitan areas and coastal regions, reflecting decades of economic development and tourism-related infrastructure growth [26]. The fine resolution of this product allows for the detection of small-scale changes that are often missed by coarser datasets, including field-level agricultural modifications, small forest clearings, and urban infill development [14].



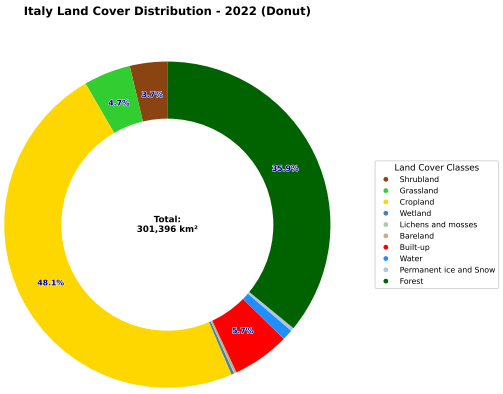

**Figure 11: Italy Land Cover Map 1985 (30m resolution)**



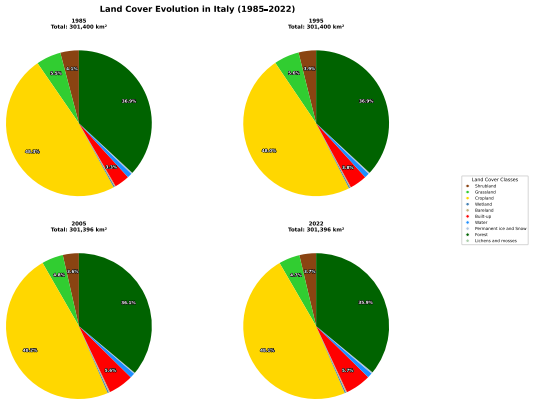

**Figure 12: Italy Land Cover Map 2022 (30m resolution)**

**Figure 13: GLC_FCS30D land cover maps of Italy at 30m resolution for (a) 1985 and (b) 2022.**

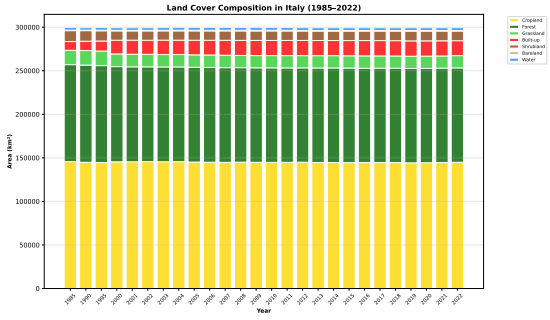
A full analysis of the GLC_FCS30D data for Italy is presented in Figures 14 and 15. The land cover statistics for 1985 and 2022 reveal the significant transformations in Italy's landscape over nearly four decades [1]. In 1985, croplands dominated with 48.32% of the national territory, followed closely by forests (36.91%) and grasslands (5.49%). By 2022, croplands had slightly decreased to 48.08% and forests had declined more noticeably to 35.94%, while built-up areas experienced the most substantial expansion, nearly doubling from 3.34% to 5.68%. These results were computed using `GRASS GIS` with per-pixel area calculations 900 $m^2$ pixel size for 30 m resolution data and a custom `Python` workflow for data manipulation and statistical analysis [37].



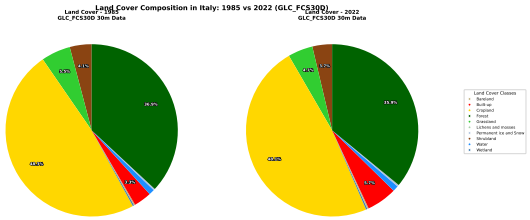

(a) Italy Land Cover Distribution (2022)




(b) Multi-year Land Cover Trends




(c) Land Cover Composition (1985 to 2022)




(d) Land Cover Distribution (1985 vs. 2022)

**Figure 14: GLC_FCS30D land cover analysis for Italy showing (a) 2022 distribution, (b) multi-year trends, (c) temporal composition, and (d) 1985-2022 comparison.**

Land Cover Statistics 1995

| | Year | Class | Label | Pixel_Count | Area_km2 | Area_m2 | Pixel_Area_m2 | Percentage |
|---|---|---|---|---|---|---|---|---|
| 0 | 1995 | 5 | Shrubland | 13,023,678 | 11,721.31 | 11,721,310,200.00 | 900.00 | 3.89 |
| 1 | 1995 | 7 | Grassland | 18,690,087 | 16,821.08 | 16,821,078,300.00 | 900.00 | 5.58 |
| 2 | 1995 | 8 | Cropland | 160,671,069 | 144,603.96 | 144,603,962,100.00 | 900.00 | 47.98 |
| 3 | 1995 | 9 | Wetland | 882,306 | 794.08 | 794,075,400.00 | 900.00 | 0.26 |
| 4 | 1995 | 12 | Bareland | 721,714 | 649.54 | 649,542,600.00 | 900.00 | 0.22 |
| 5 | 1995 | 13 | Built-up | 12,781,494 | 11,503.34 | 11,503,344,600.00 | 900.00 | 3.82 |
| 6 | 1995 | 15 | Water | 3,568,619 | 3,211.76 | 3,211,757,100.00 | 900.00 | 1.07 |
| 7 | 1995 | 16 | Permanent ice and Snow | 1,010,623 | 909.56 | 909,560,700.00 | 900.00 | 0.30 |
| 8 | 1995 | 20 | Forest | 123,538,782 | 111,184.90 | 111,184,903,800.00 | 900.00 | 36.89 |

(a) Land Cover Statistics (1995)

Land Cover Statistics 2022

| | Year | Class | Label | Pixel_Count | Area_km2 | Area_m2 | Pixel_Area_m2 | Percentage |
|---|---|---|---|---|---|---|---|---|
| 0 | 2022 | 5 | Shrubland | 12,404,204 | 11,163.78 | 11,163,783,600.00 | 900.00 | 3.70 |
| 1 | 2022 | 7 | Grassland | 15,612,186 | 14,050.97 | 14,050,967,400.00 | 900.00 | 4.66 |
| 2 | 2022 | 8 | Cropland | 161,002,186 | 144,901.97 | 144,901,967,400.00 | 900.00 | 48.08 |
| 3 | 2022 | 9 | Wetland | 907,565 | 816.81 | 816,808,500.00 | 900.00 | 0.27 |
| 4 | 2022 | 11 | Lichens and mosses | 148,757 | 133.88 | 133,881,300.00 | 900.00 | 0.04 |
| 5 | 2022 | 12 | Bareland | 712,678 | 641.41 | 641,410,200.00 | 900.00 | 0.21 |
| 6 | 2022 | 13 | Built-up | 19,023,191 | 17,120.87 | 17,120,871,900.00 | 900.00 | 5.68 |
| 7 | 2022 | 15 | Water | 3,613,757 | 3,252.38 | 3,252,381,300.00 | 900.00 | 1.08 |
| 8 | 2022 | 16 | Permanent ice and Snow | 1,108,687 | 997.82 | 997,818,300.00 | 900.00 | 0.33 |
| 9 | 2022 | 20 | Forest | 120,351,297 | 108,316.17 | 108,316,167,300.00 | 900.00 | 35.94 |

(b) Land Cover Statistics (2022)

**Figure 15: Comparative land cover statistics from the GLC_FCS30D dataset showing (a) 1995 and (b) 2022 distributions.**

In conclusion, the integrated multi-source analysis reveals two unequivocal and major trends defining Italy's recent land cover history: significant urban expansion, primarily at the expense of agricultural land, and a associated, though less severe, contraction of

both agricultural and forest land. The dynamics of natural and semi-natural vegetation (grassland, shrubland) remain less certain and are highly sensitive to the observation scale and classification system. This ambiguity underscores the critical importance of employing a multi-source analytical approach to achieve a nuanced and robust understanding of complex land cover change processes.

## 1.11. Land Cover Transition Analysis

This section presents a comprehensive analysis of land cover transitions in Italy, quantifying the pathways, magnitude, and dynamics of landscape change from 1985 to 2022. Utilizing a dual-resolution approach with the GLC_FCS30D (30m) and ESA CCI LC (300m) datasets, the analysis reveals the complex interplay of urbanization, agricultural change, and forest dynamics that happened in the Italian peninsula [38].

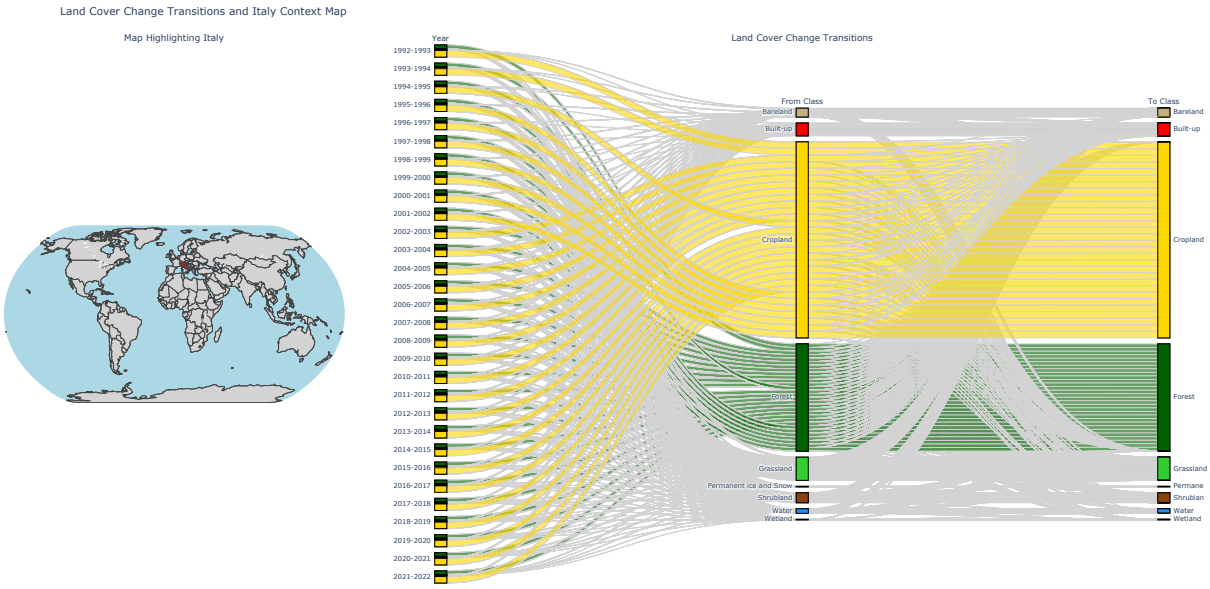
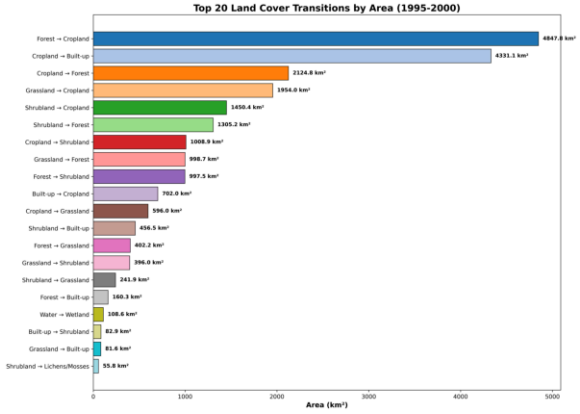
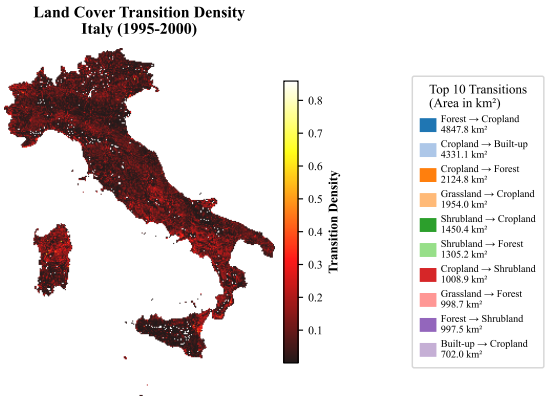



**Figure 16: Land Cover Transition in Italy showing complex pathways of change between major land cover classes.**




(a) Top land cover transitions by area (1995–2000).




(b) Land Cover Transition Density in Italy (1995-2000). Areas with warmer colors indicate higher occurrence of land cover changes.

**Figure 17: Short-term vs. long-term land cover dynamics. (a–b) illustrate annual-scale transitions between 1995 and 2000, highlighting local variability and dominant processes in this period. (c) summarizes the top 10 transition pathways across the entire study period (1985–2022), capturing long-term structural changes.**

The high-resolution GLC_FCS30D analysis reveals a landscape of remarkable dynamism, dominated by three key processes: urbanization, agricultural change, and forest dynamics. The top 10 transition pathways are summarized in Table 6.

**Table 6: Major land cover transitions from GLC_FCS30D analysis (1985–2022). The table highlights the cumulative top 10 pathways, providing a long-term context to the short-term patterns shown in Figure 17.**

| Transition Pathway | Total Area (km²) | Percentage of Total Change | Primary Periods |
|---|---|---|---|
| Cropland Built-up | 21,560.77 | 15.3% | 2000–2010, 2015–2022 |
| Forest Cropland | 20,720.49 | 14.7% | 1995–2000, 2005–2010 |
| Cropland Forest | 17,181.29 | 12.2% | 2010–2022 |
| Built-up Cropland | 15,182.53 | 10.8% | 2000–2010 |
| Grassland Cropland | 8,196.69 | 5.8% | 2000–2010 |
| Shrubland Cropland | 8,173.27 | 5.8% | 1995–2005 |
| Cropland Shrubland | 7,383.70 | 5.2% | 2010–2022 |
| Grassland Built-up | 6,692.12 | 4.8% | 2005–2015 |
| Shrubland Forest | 5,399.88 | 3.8% | 2010–2022 |
| Forest Shrubland | 5,332.83 | 3.8% | 1995–2005, 2010–2015 |

The land cover transition analysis was conducted using two complementary methodological approaches to quantify and characterize landscape changes across Italy. The analysis employed GRASS GIS for robust geospatial processing, leveraging custom Python scripts designed for multi-resolution transition detection [39]. The high-resolution analysis utilized 30m GLC_FCS30D data, comprising 26 annual land cover maps from 1985 to 2022. This processing workflow included pixel-level transition computation using `r.mapcalc` with combined class coding, exact pixel area calculation (approximately 900 $m^2$ per pixel) for area quantification, an annual consecutive pair analysis that generated 25 transition matrices, and the production of metadata-enriched GeoTIFF exports with compression and tiling optimization. Concurrently, a medium-resolution analysis employed 300m ESA CCI LC data with 31 annual maps from 1992 to 2022. This methodology featured 300m-optimized processing with pixel area validation (∼90,000 $m^2$ per pixel), an annual consecutive pair analysis generating 30 transition matrices, and resolution-specific parameterization to account for coarser spatial detail, all within a comparative framework aligned with the high-resolution analysis. Both methodologies implemented quality control, including projection verification (EPSG:3035) and pixel area validation. The total computational workflow processed 56 transition periods across both datasets, generating detailed transition matrices and a combined analysis ready for statistical evaluation [38]. The high-resolution GLC_FCS30D dataset revealed intricate land cover dynamics across Italy over 37 years (Table 6), detecting a total change area of 140,874.24 $km^2$ (excluding stable areas) and demonstrating remarkable landscape dynamism with several dominant conversion pathways [40].

Urbanization emerged as a dominant process, characterized by cropland-to-built-up conversions (21,560.77 $km^2$) concentrated in 2000–2010 and 2015–2022. A particularly significant finding was the detection of substantial built-up-to-cropland transitions (15,182.53 $km^2$), suggesting intricate land recycling, redevelopment, and peri-urban transformation processes that are only visible at high resolution [41].

Agricultural transitions showed significant forest-to-cropland conversion (20,720.49 $km^2$) during 1995–2000 and 2005–2010, indicating periods of agricultural expansion. Conversely, substantial cropland-to-forest transitions (17,181.29 $km^2$) after 2010 suggested widespread agricultural abandonment and subsequent reforestation, particularly in marginal agricultural areas [42]. The analysis also revealed extensive exchanges between cropland and natural vegetation (grassland and shrubland), indicating a highly dynamic agricultural frontier.

Forest dynamics exhibited both degradation and recovery patterns, with shrubland serving as a transitional class between forest and cropland. The significant shrubland-

to-forest transitions (5,399.88 km$^2$) and forest-to-shrubland transitions (5,332.83 km$^2$) indicated active successional processes and management interventions across Italian landscapes [43].

The ESA CCI LC dataset captured broader-scale patterns while revealing a fundamental finding: it detected approximately 6.25 times less total change area than the high-resolution data. This highlights the substantial underestimation of land change inherent in coarser resolution data [44]. The dominant pathways it did capture, primarily large-scale forest-cropland exchanges and urbanization from bare surfaces, are summarized in Table 7.

**Table 7: Major land cover transitions from ESA CCI LC analysis (1992–2022)**

| Transition Pathway | Total Area (km$^2$) | Percentage of Total Change | Primary Periods |
|---|---|---|---|
| Forest Cropland | 5,979.15 | 26.5% | 1994–2000 |
| Cropland Forest | 5,416.47 | 24.0% | 2010–2022 |
| Cropland Built-up | 3,488.22 | 15.5% | 2000–2010 |
| Bareland Built-up | 2,981.61 | 13.2% | 1992–2005 |
| Forest Grassland | 1,718.10 | 7.6% | 1998–2008 |
| Forest Shrubland | 992.34 | 4.4% | 1998–2005 |
| Grassland Forest | 489.69 | 2.2% | 2005–2015 |
| Forest Built-up | 324.63 | 1.4% | 2000–2010 |
| Shrubland Forest | 299.07 | 1.3% | 2005–2015 |
| Forest Bareland | 242.37 | 1.1% | 1994–2005 |

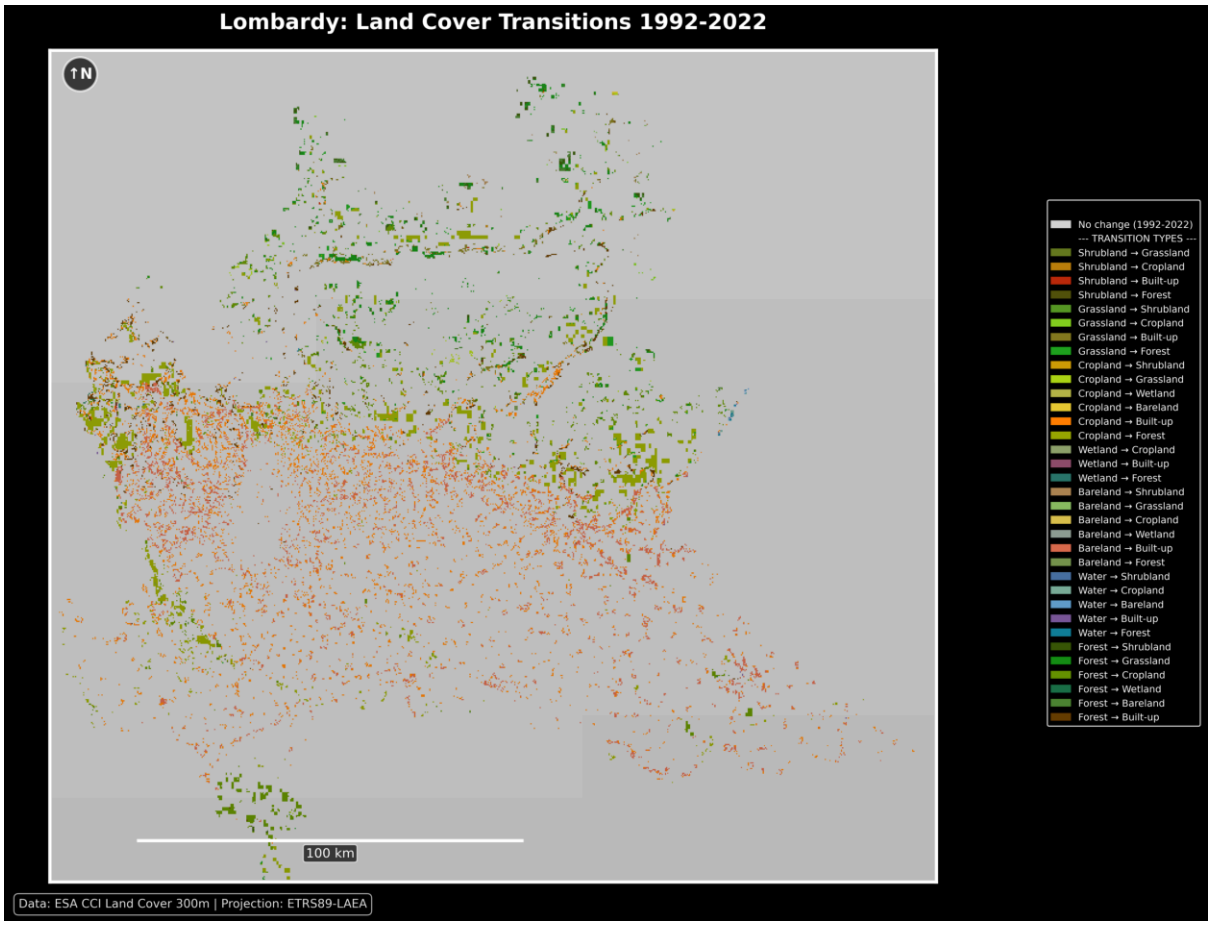



**Figure 18: Land cover transitions in Lombardy, Italy (1992–2022). The map visualizes thirty years of land cover change derived from ESA CCI 300m resolution data, showing conversion pathways between major land cover classes. Colored areas represent transitions between different land cover types using a blended color scheme (e.g., forest-to-cropland transitions appear as orange hues, while cropland-to-forest regrowth shows as green tones). Light gray areas indicate stable land cover with no change over the study period. The black outline delineates the Lombardy regional boundary, providing spatial context for one of Italy's most economically significant and dynamically changing regions.**

This dataset highlighted forest-to-cropland transitions as the dominant process (5,979.15 $km^2$, 26.5%), particularly during 1994–2000, followed by substantial cropland-to-forest recovery after 2010 (5,416.47 $km^2$, 24.0%). The relative importance of these transitions was more pronounced in the medium-resolution data compared to the high-resolution analysis due to the significantly smaller denominator of total detected change.

Urbanization patterns revealed different pathways compared to the high-resolution data, with significant bareland-to-built-up transitions (2,981.61 $km^2$, 13.2%) that were

not prominent in the GLC_FCS30D analysis. This suggests the 300m resolution may conflate certain land cover types or detect different aspects of urban expansion processes, particularly the conversion of large, contiguous non-vegetated areas.

The analysis of land cover transitions reveals a landscape undergoing significant and complex change[45]. The sheer magnitude of gross changes the total area that experienced a land cover conversion far exceeds the net changes, indicating a highly dynamic system with substantial simultaneous losses and gains across most classes [46].

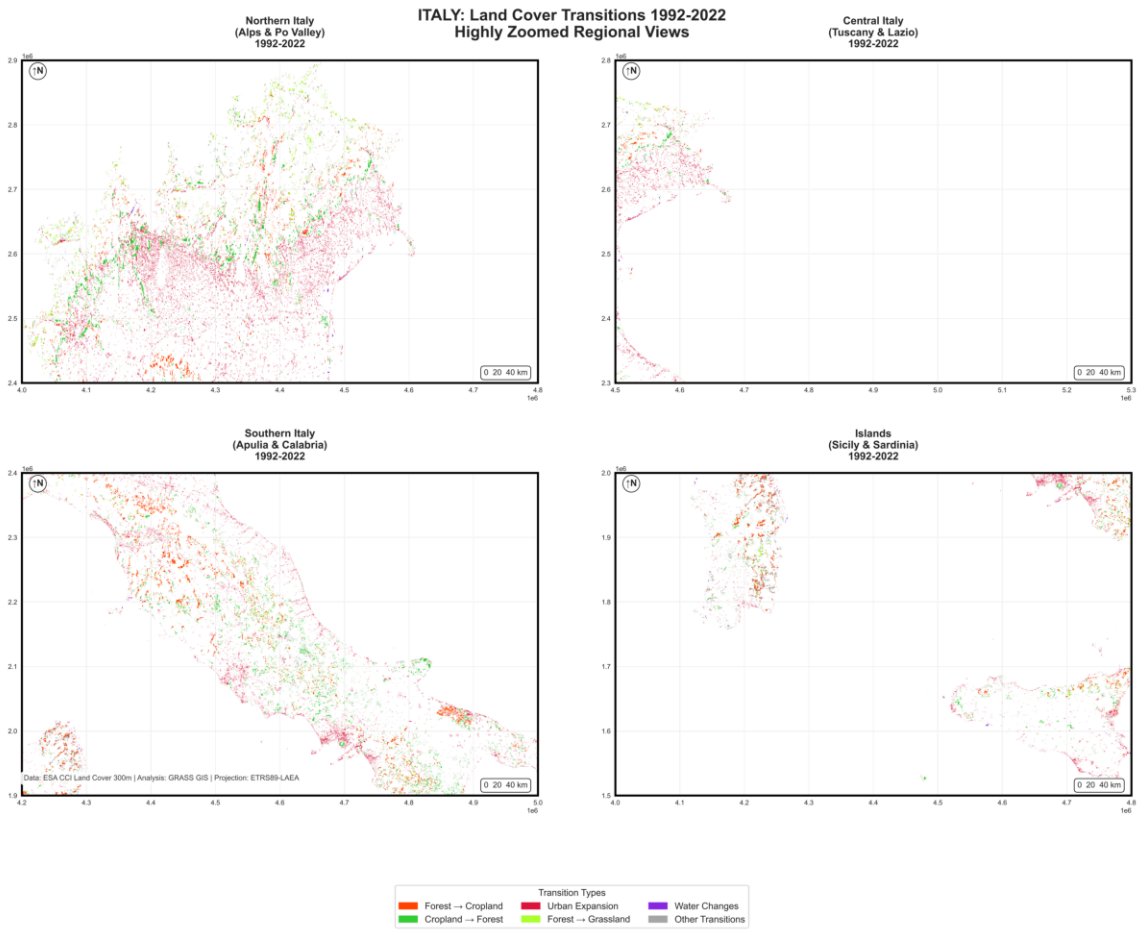



**Figure 19: Land cover transitions in Italy (1992–2022) showing regional patterns of change, including urban expansion (red), forest-cropland conversions (orange/green), and other major land use transformations across four key geographical regions.**

Figure 20 visualizes the major land cover fluxes during the initial period of the study (1985–2006), capturing the interchange between dominant classes. The most prominent flows are consistently between **Cropland**, **Forest**, and **Shrubland**, illustrating a pat-

tern of agricultural expansion, abandonment, and subsequent natural regrowth. The emergence of **Built-up** areas as a significant destination class is also visibly clear, marking the ongoing process of urbanization.

Figure 21 shows the transition pathways for the 1992-2022 period from the ESA CCI LC data, revealing different patterns dominated by Forest-Cropland exchanges and Bareland to Built-up transitions.

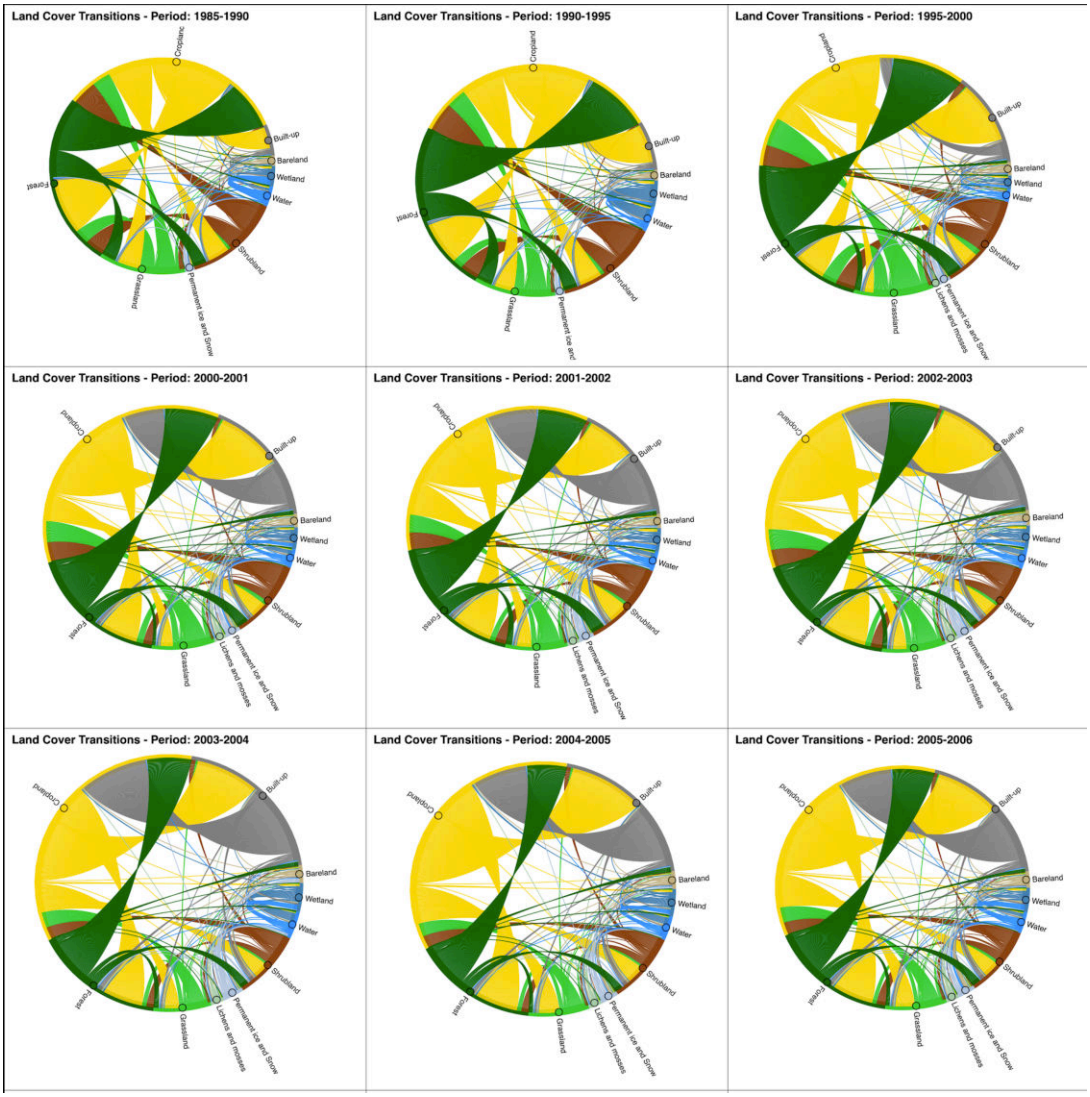



**Figure 20: Major land cover transition pathways for the period 1985–2006. The width of each chord is proportional to the area ($km^2$) converted from one class to another. The diagram highlights the high dynamism and key exchanges between Cropland, Forest, and Shrubland, alongside the growing influence of Built-up areas.**

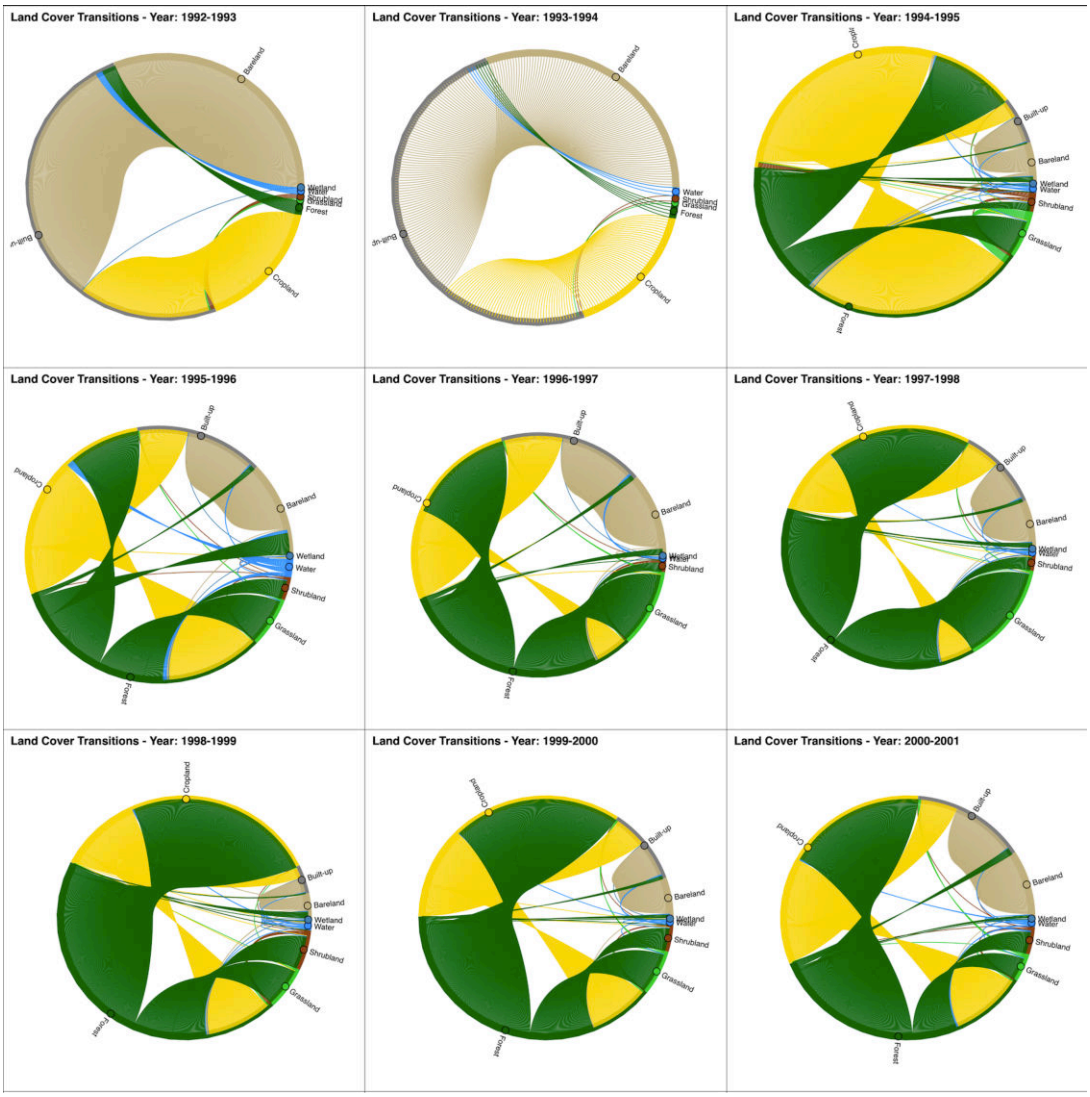
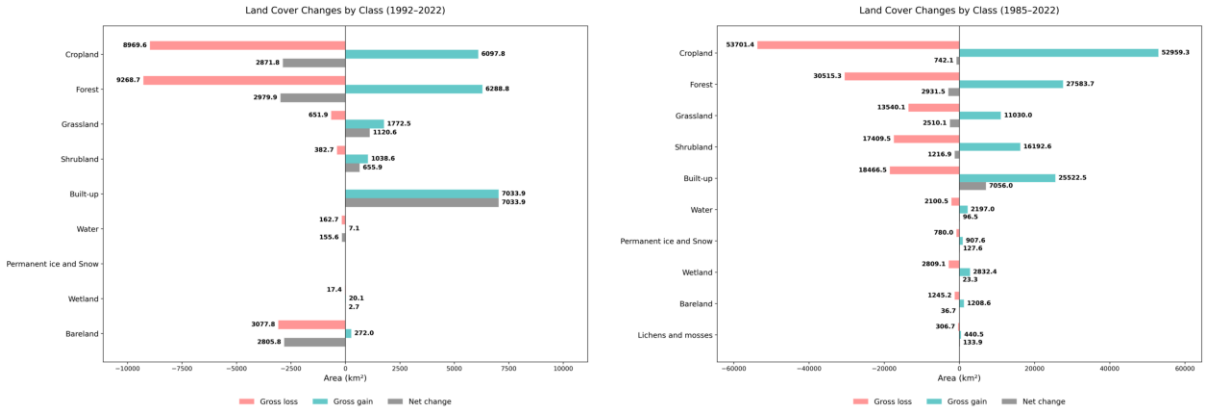



**Figure 21: Major land cover transition pathways in Italy (1992–2022) from ESA CCI LC data. The chord diagram visualizes dominant transitions including Forest-Cropland exchanges (26.5% and 24.0% of total change), urban expansion from Cropland (15.5%) and Bareland (13.2%), and Forest-Grassland conversions (7.6%).**

The multi-temporal analysis reveals that the observed scale and complexity of land cover change are dependent on the spatial resolution of the source data. This is demonstrated by a comparison of the GLC-FCS30D product (30 m) with the ESA-CCI product (300 m).

The GLC-FCS30D dataset covers a longer time span (1985–2022) than the ESA-CCI dataset (1992–2022). The results indicate that the finer spatial resolution of the GLC-FCS30D data is a primary factor in its detection of a greater magnitude and variety of changes, while the extended temporal coverage contributes additional change area.

This difference is shown in Figure 22 and quantified in Table 8.




**(a) 1992–2022 Period (ESA CCI)** **(b) 1985–2022 Period (GLC_FCS30D)**

**Figure 22: Comparative horizontal bar plots showing gross losses, gross gains, and net changes in land cover classes in Italy across two time periods and datasets. Values represent area in km².**

**Table 8: Comparative Land Cover Change Results (km²)**

| | 1992–2022 (ESA CCI) | | | 1985–2022 (GLC_FCS30D) | | |
|---|---|---|---|---|---|---|
| **Land Cover Class** | **Net Change (km²)** | **Gross Loss (km²)** | **Gross Gain (km²)** | **Net Change (km²)** | **Gross Loss (km²)** | **Gross Gain (km²)** |
| Cropland | -2,871.8 | 8,969.6 | 6,097.8 | -742.1 | 53,701.4 | 52,959.3 |
| Forest | -2,979.9 | 9,268.7 | 6,288.8 | -2,931.5 | 30,515.3 | 27,583.7 |
| Grassland | 1,120.6 | 651.9 | 1,772.5 | -2,510.1 | 13,540.1 | 11,030.0 |
| Shrubland | 655.9 | 382.7 | 1,038.6 | -1,216.9 | 17,409.5 | 16,192.6 |
| Built-up | 7,034.0 | 0.0 | 7,034.0 | 7,056.0 | 18,466.5 | 25,522.5 |
| Water | -155.6 | 162.7 | 7.1 | 96.5 | 2,100.5 | 2,197.0 |
| Perm. ice and Snow | 0.0 | 0.0 | 0.0 | 127.6 | 780.0 | 907.6 |
| Wetland | 2.7 | 17.4 | 20.1 | 23.3 | 2,809.1 | 2,832.4 |
| Bareland | -2,805.8 | 3,077.8 | 272.0 | -36.7 | 1,245.2 | 1,208.6 |
| Lichens and mosses | – | – | – | 133.9 | 306.7 | 440.5 |

The differences in the observed magnitude and type of land change are a consequence of the technical characteristics of each product, namely the spatial resolution (30m vs. 300m) and the temporal coverage (annual vs. 5-yearly intervals in the early period).

The GLC_FCS30D dataset (30m; 1985–2022) shows substantial growth in **built-up ar-**

**eas** (+7,056.0 ), characterized by gross changes (18,466.5  losses and 25,522.5  gains) that suggest urban land recycling and redevelopment [41]. It also measured extensive gross dynamics in **forest areas** (30,515.3  losses and 27,583.7  gains) leading to a net decline (-2,931.5 ), which can indicate processes like selective logging and small-scale clearing [47]. **Cropland** transformations were also more extensive in this dataset (53,701.4  gross loss), suggesting a pattern of abandonment and intensification [42].

The ESA CCI dataset (300m; 1992–2022) captured a net decline in **forest** and **cropland**, but the magnitude of gross changes was lower. This is consistent with the understanding that coarser resolution data can underestimate total change flux by blending fine changes within large pixels [44]. The differences in results for classes like **grassland** and **shrubland**, which show opposite net trends, are likely artifacts of the combined effects of resolution and classification schema differences, where spectral mixing within large pixels can misrepresent heterogeneous landscapes [48, 49].

The greater gross changes observed over the 1985–2022 period are therefore attributed to the higher spatial resolution of the input data, which detects fine changes, with the extended temporal coverage contributing additional change area.

The GLC_FCS30D dataset captured 2,440 unique transitions compared to 926 in the ESA CCI dataset. This suggests that finer spatial resolution can detect a wider array of processes. The results indicate that higher resolution data may be better suited for detecting fine-scale, bidirectional exchanges, while medium-resolution data tends to capture the net outcome of larger conversions [38]. This analysis shows that the spatial resolution of the source data influences the understanding of landscape processes [44].

A direct comparison of the two datasets (Table 9) shows that the spatial resolution of the data influences which processes are detected. The empty cells (‘-‘) indicate transitions that were not captured by a given dataset.

**Table 9: Comparison of major land cover transitions: GLC_FCS30D (1985–2022) vs. ESA CCI LC (1992–2022)**

| Transition Pathway | GLC_FCS30D Area (km²) | GLC % | ESA CCI Area (km²) | ESA % |
|---|---|---|---|---|
| Cropland Built-up | 21,560.77 | 15.3% | 3,488.22 | 15.5% |
| Forest Cropland | 20,720.49 | 14.7% | 5,979.15 | 26.5% |
| Cropland Forest | 17,181.29 | 12.2% | 5,416.47 | 24.0% |
| Built-up Cropland | 15,182.53 | 10.8% | – | – |
| Grassland Cropland | 8,196.69 | 5.8% | – | – |
| Shrubland Cropland | 8,173.27 | 5.8% | – | – |
| Bareland Built-up | – | – | 2,981.61 | 13.2% |
| Forest Grassland | – | – | 1,718.10 | 7.6% |
| Forest Shrubland | 5,332.83 | 3.8% | 992.34 | 4.4% |
| Shrubland Forest | 5,399.88 | 3.8% | 299.07 | 1.3% |

Several transitions were not detected by ESA CCI, including conversions from Built-up to Cropland (15,182 ), Grassland to Cropland (8,196 ), and Shrubland to Cropland (8,173 ). These transitions often occur as fine-scale changes at landscape interfaces that can be obscured within the 300m pixels of the ESA dataset due to mixed-pixel effects [21]. The ESA CCI dataset detected transitions such as Bareland to Built-up (2,981 ) and Forest to Grassland (1,718 ), which represent larger and more homogeneous conversions that are more readily captured at coarser resolutions [11]. This shows that the spatial resolution of analysis influences the observed land system dynamics [40].

Both datasets showed forest cropland exchanges as important, though they diverged in relative ranking. The proportional representation of forest-related transitions was higher in the coarser-resolution dataset due to its smaller total detected change area. These differences highlight methodological considerations for land change studies. **Resolution effects** are a factor, as the GLC_FCS30D dataset (30m) detected finer-scale transitions that the ESA CCI LC dataset (300m) did not capture [44]. The **minimum mapping unit** of ESA CCI LC can simplify transition pathways and lead to loss of detail in heterogeneous landscapes [21]. **Classification schema** differences also contribute to variation, particularly for mixed vegetation classes and heterogeneous urban areas [50].

The multi-scale analysis provides different insights for land system science and management. The data on complex urbanrural dynamics such as land recycling (Built-up to Cropland) may be relevant for **urban planning** strategies for peri-urban interfaces [41]. The prominence of forest cropland exchanges highlights ongoing pressures on ecosys-

tems, which may be of interest for **forest management** and integrated forest agriculture policies [47]. For **agricultural policy**, the bidirectional nature of transitions indicates complex socioeconomic drivers [42]. For **monitoring systems**, the results show that data resolution is an important consideration for study design [40].

Analysis of temporal patterns showed distinct phases of land cover change in Italy [38]. The early period (1985–2000) was characterized by moderate changes involving gradual urbanization and agricultural intensification, while forest cover remained relatively stable [48]. The mid-period (2000–2010) showed accelerated urbanization, with built-up areas expanding and increased transitions between agricultural and natural vegetation classes [41]. The late period (2010–2022) showed a stabilization of some processes, with continued but slower urban growth, and a moderation in forest decline [47].

Spatial analysis showed regional heterogeneity [40]. Northern Italy experienced the most intensive urbanization, particularly in the Po Valley, alongside agricultural declines, while Alpine regions remained relatively stable. Central Italy exhibited agricultural abandonment and secondary succession. Southern Italy and the islands showed more variable patterns, with intensification and abandonment occurring simultaneously [38].

The workflow incorporated a **data processing pipeline** to harmonize raw class codes into standardized categories across the 37-year study period [46]. It included **chronological sequencing** algorithms to order transition periods, yielding 25 consecutive 5-year periods from 1985–1990 through 2021–2022 [48]. **Metrics** were calculated, including total transition area, annual rate of land cover change intensity, class-specific transitions, and net change dynamics [18].

Annual rate of Land Cover change intensity for each period was calculated as

$$I_p = \frac{\Delta A_p}{T_p}, \tag{15}$$

where $I_p$ is the annual rate of change intensity for period $p$ (/year), $\Delta A_p$ is the total area changed in period $p$ (), and $T_p$ is the duration of period $p$ (years). The analysis identified 1995–2000 as a period with significant transitions, with 23,170  of total change and an annual intensity of 3,862 /year. Overall, 2,440 transitions were recorded, covering a cumulative 140,874  across the study period [38].

This temporal sequencing showed phases of agricultural intensification, urban expansion, and vegetation dynamics, with the mid-to-late 1990s as a period for further investigation into underlying drivers [43]. The analysis provides a framework for long-term transitions [38], identifies periods of transformation, enables comparative cross-scale stud-

ies [40], and may support land management strategies based on temporal dynamics [47].

The hotspot analysis (Figure 26) identifies the mid-to-late 1990s as a period of landscape transformation.

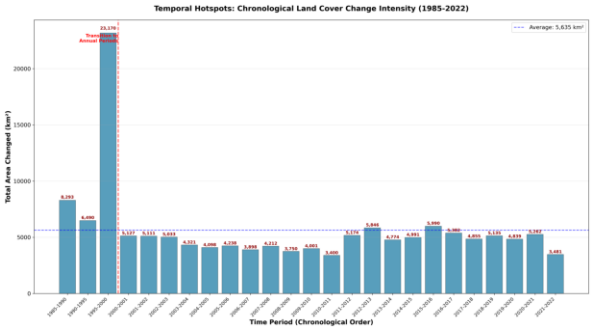



**Figure 23: Chronological analysis of total transition area.**

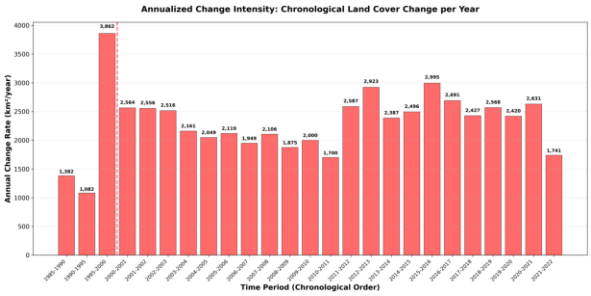



**Figure 24: Annual Rate of Land Cover Change by Period**

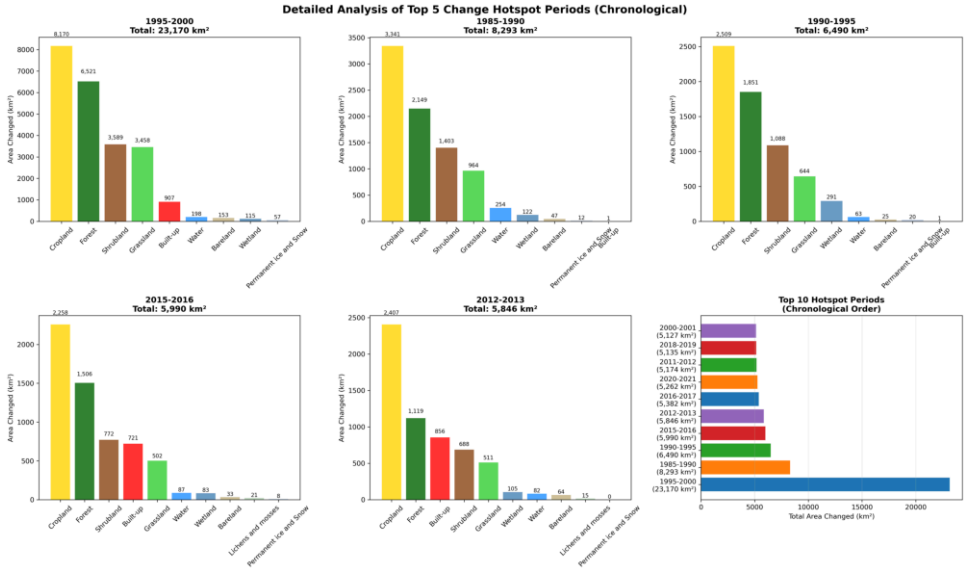



**Figure 25: Detailed breakdown of top hotspot periods and transitions.**

**Figure 26: Comprehensive temporal hotspot analysis of land cover transitions in Italy (1985–2022), showing the chronological sequence of changes (top left), the annualized intensity of those changes (top right), and a detailed breakdown of the most significant transitions (bottom).**

## 1.12. Accuracy Assessment

This section presents accuracy assessment between the high-resolution GLC_FCS30D (30m) and medium-resolution ESA CCI LC (300m) land cover datasets across Italy. The period 2000 to 2022 was considered in this analysis. The evaluation provides insights into

the consistency, reliability, and limitations of both datasets for land cover monitoring applications, revealing both substantial agreement and systematic discrepancies that varied significantly by land cover class [12, 14, 50].

The accuracy assessment employed a methodological framework implemented in GRASS GIS with custom Python scripting. The GLC_FCS30D data (30m) was resampled to 300m resolution using a majority filter to match the spatial resolution of the ESA CCI LC dataset, enabling direct pixel-to-pixel comparison. For each year from 2000 to 2022, comprehensive cross-tabulation matrices were generated, and standardized accuracy metrics were calculated using the scikit-learn Python library [51]. The assessment employed established statistical measures including Overall Accuracy (OA), Cohen's Kappa Coefficient ($\kappa$), Producer's Accuracy (PA), and User's Accuracy (UA). These metrics were calculated for each land cover class following the classification system shown in Table 10 [18, 52].

**Table 10: Land cover class codes and names used for accuracy assessment**

| Class Code | Land Cover Class |
|---|---|
| 5 | Shrubland |
| 7 | Grassland |
| 8 | Cropland |
| 9 | Wetland |
| 11 | Lichens/Mosses |
| 12 | Bareland |
| 13 | Built-up |
| 15 | Water |
| 16 | Ice/Snow |
| 20 | Forest |

The inter-dataset comparison revealed consistently high agreement between the datasets over the 22-year study period [13, 53]. As shown in Table 11 and visualized in Figure 27, overall accuracy ranged between 78.06% and 79.13%, demonstrating remarkable temporal stability.

**Table 11: Overall accuracy and Kappa coefficient trends (2000–2022)**

| Year | 2000 | 2002 | 2004 | 2006 | 2008 | 2010 | 2012 | 2014 | 2016 | 2018 | 2020 | 2022 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Accuracy (%)** | 78.06 | 78.41 | 78.68 | 78.86 | 79.13 | 79.12 | 79.13 | 79.03 | 79.03 | 78.95 | 78.75 | 78.77 |
| **Kappa ()** | 0.639 | 0.645 | 0.651 | 0.654 | 0.659 | 0.659 | 0.660 | 0.659 | 0.659 | 0.657 | 0.654 | 0.654 |

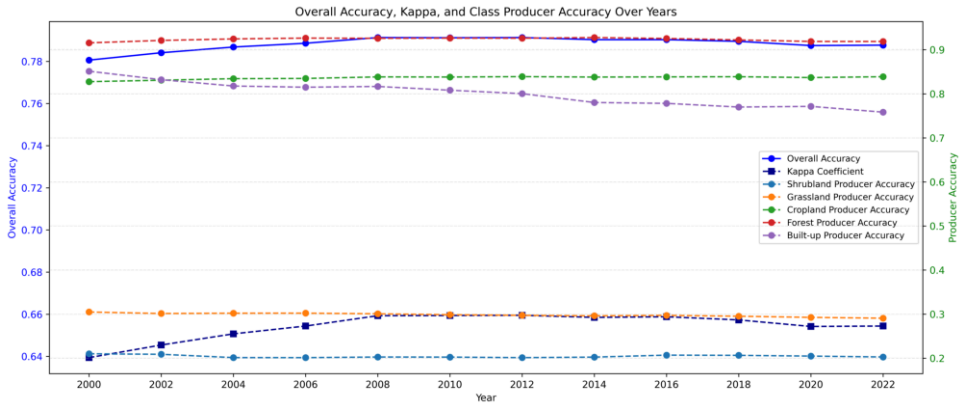



**Figure 27: Temporal trends in overall accuracy and Kappa coefficient (2000–2022). The remarkable consistency demonstrates stable performance of both classification systems despite land cover changes over the 22-year period.**

The Kappa values, ranging from 0.639 to 0.660, indicate substantial agreement according to Landis and Koch's interpretation guidelines [54]. A slight increasing trend in accuracy is observable from 2000 to 2012, followed by relative stability with minor fluctuations in later years [55, 56]. This temporal consistency suggests that both classification systems maintained stable performance despite landscape changes over the two-decade period.

The assessment revealed significant variations in agreement across different land cover types, as comprehensively detailed in Table 12 and visualized in Figures 28 and 29 [40, 57].

**Table 12: Class-wise accuracy patterns (average 2000–2022)**

| Land Cover Class | Producer's Accuracy (%) | User's Accuracy (%) | Agreement Level |
|---|---|---|---|
| Forest | 92.21 | 71.55 | High |
| Cropland | 83.49 | 88.72 | High |
| Water | 73.12 | 89.45 | High |
| Ice/Snow | 87.68 | 32.27 | Moderate |
| Built-up | 80.43 | 72.31 | Moderate |
| Wetland | 44.01 | 24.95 | Low |
| Grassland | 29.92 | 54.71 | Low |
| Bareland | 4.73 | 70.57 | Low |
| Shrubland | 20.45 | 26.07 | Low |
| Lichens/Mosses | 0.00 | 0.00 | Very Low |

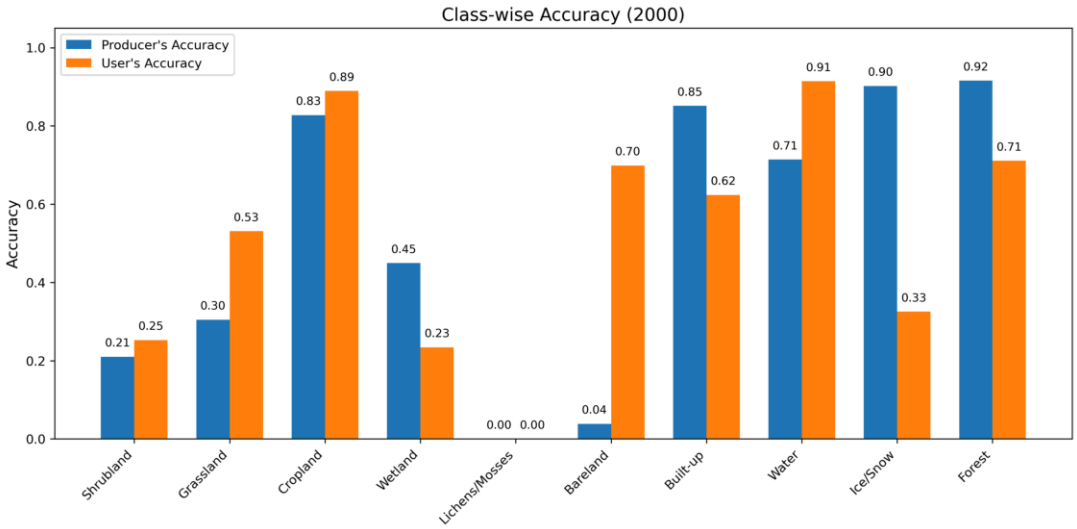



**Figure 28: Class-wise producer's and user's accuracy for 2000, showing the foundational patterns of agreement and disagreement that persisted throughout the study period.**

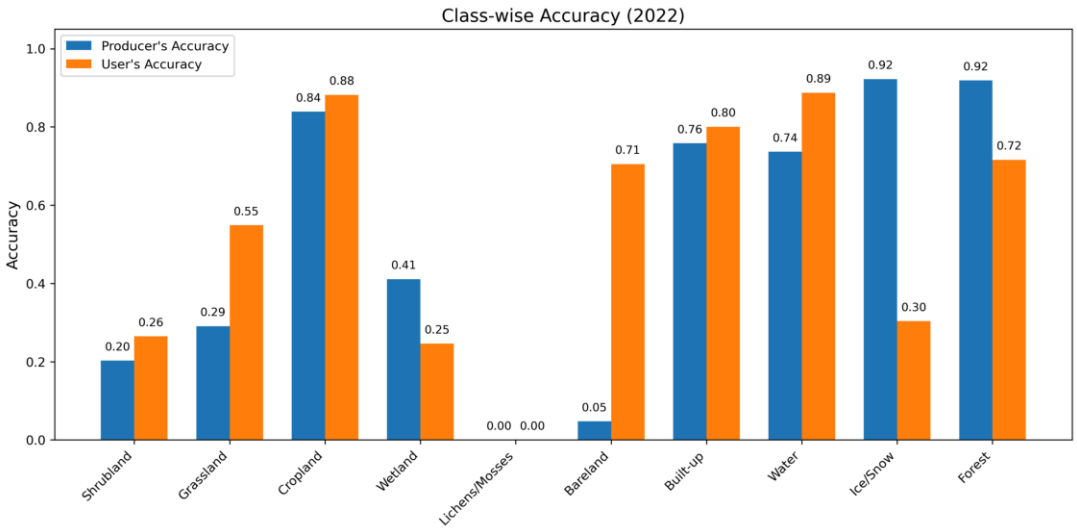



**Figure 29: Class-wise producer's and user's accuracy for 2022, demonstrating the remarkable stability of accuracy patterns over the 22-year assessment period.**

The class-wise analysis revealed three distinct tiers of agreement. **High-Agreement Classes: Forest** exhibited excellent Producer's Accuracy (92.21%) but moderate User's Accuracy (71.55%), indicating that while most forest areas were correctly identified, the class frequently included other vegetated areas [11]. **Cropland** showed balanced high accuracy (PA: 83.49%, UA: 88.72%), suggesting reliable identification of agricultural areas [58]. **Water** demonstrated strong User's Accuracy (89.45%) with good Producer's Accuracy (73.12%), indicating high confidence in water identification [59]. **Moderate-Agreement Classes: Ice/Snow** showed a characteristic pattern of high Producer's Accuracy (87.68%) but very low User's Accuracy (32.27%), indicating that while most snow/ice areas were correctly identified, the class might be frequently misapplied to other bright surfaces Paul et al. [60] . **Built-up** areas displayed moderate accuracy levels (PA: 80.43%, UA: 72.31%), with confusion primarily occurring with bareland and some agricultural areas [45]. **Low-Agreement Classes:** The vegetation classes **Shrubland** and **Grassland** showed substantial confusion, reflecting the spectral similarity of these vegetation types and the challenges of differentiating them at medium resolution [61]. **Wetlands** showed low accuracy for both metrics, with confusion distributed across water and vegetated classes [62]. **Bareland** exhibited extremely low Producer's Accuracy, indicating severe omission errors [63]. **Lichens/Mosses** showed complete disagreement, suggesting fundamental differences in how this class is defined or detected between the two datasets.

Analysis of confusion matrices revealed consistent patterns of inter-class confusion that

persisted throughout the study period [44, 64] . Figures 30 and 31 visualize these patterns for the beginning and end of the study period.

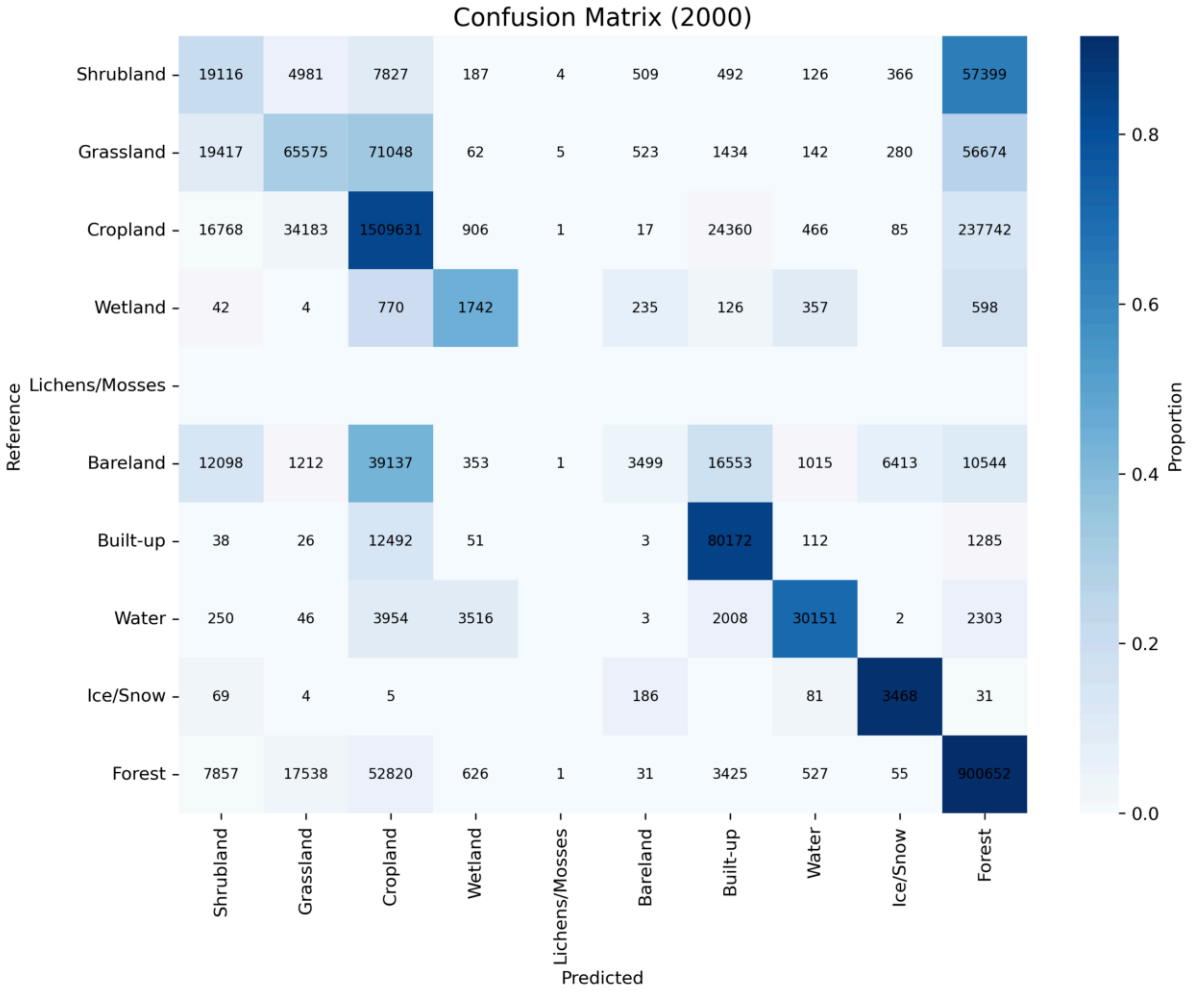



**Figure 30: Normalized confusion matrix for 2000 showing persistent confusion patterns. The strong diagonal indicates general agreement, while off-diagonal elements reveal systematic confusions, particularly among vegetation classes and between built-up and bareland.**

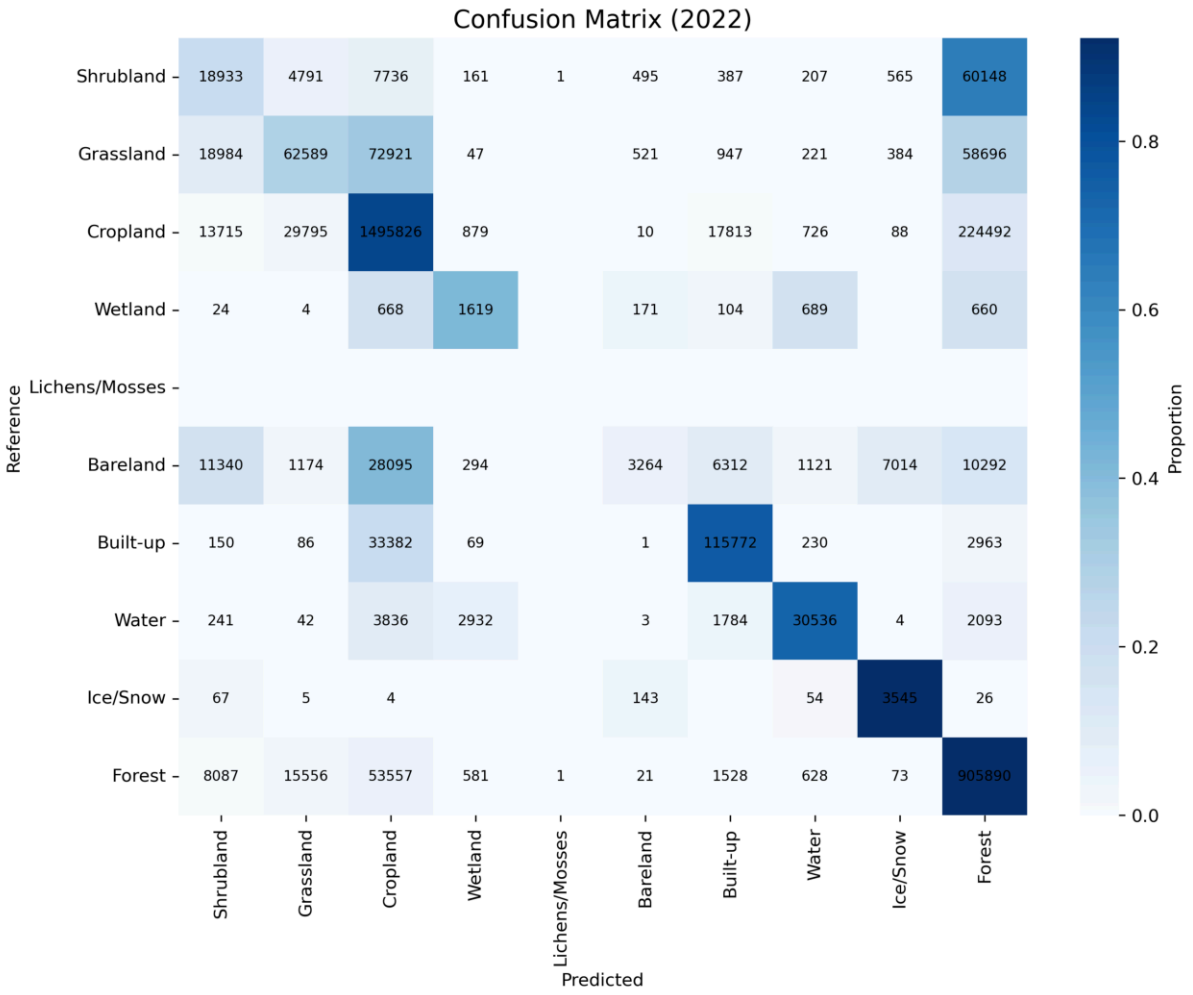
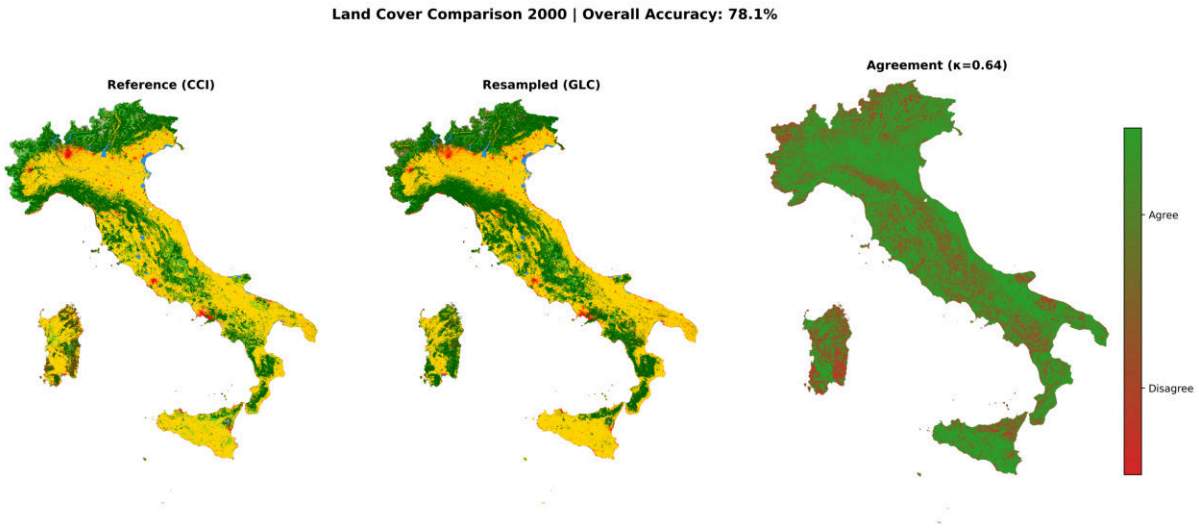


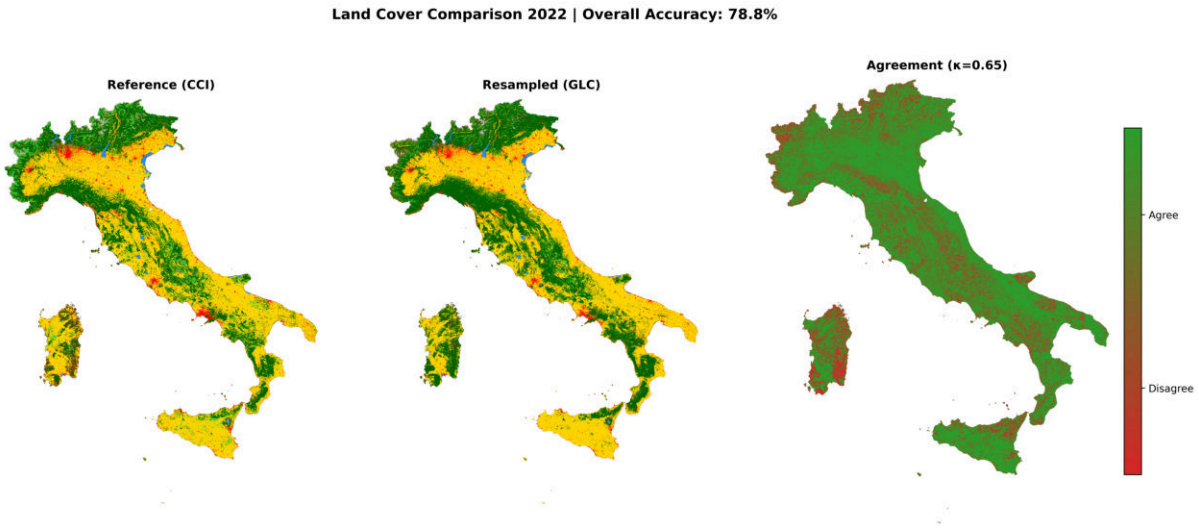

**Figure 31: Normalized confusion matrix for 2022 demonstrating the stability of confusion patterns over time. The consistent off-diagonal patterns indicate systematic rather than random errors in classification agreement.**

The spatial distribution of agreement between datasets revealed distinct geographical patterns [6, 40], as shown in Figures 32 and 33.




| Land Cover Class | Producer's Accuracy (%) | User's Accuracy (%) |
|---|---|---|
| Shrubland | 21.0 | 25.3 |
| Grassland | 30.5 | 53.1 |
| Cropland | 82.8 | 88.9 |
| Wetland | 45.0 | 23.4 |
| Lichens and mosses | 0.0 | 0.0 |
| Bareland | 3.9 | 69.9 |
| Built-up | 85.1 | 62.4 |
| Water | 71.4 | 91.4 |
| Permanent Ice and Snow | 90.2 | 32.5 |
| Forest | 91.6 | 71.1 |

**Figure 32: Spatial pattern of agreement between datasets for 2000. High agreement (green) predominates in homogeneous agricultural plains and large forest patches, while disagreement (red) concentrates in ecotones, mountainous regions, and urban peripheries.**




| Land Cover Class | Producer's Accuracy (%) | User's Accuracy (%) |
|---|---|---|
| Shrubland | 20.3 | 26.5 |
| Grassland | 29.1 | 54.9 |
| Cropland | 83.9 | 88.2 |
| Wetland | 41.1 | 24.6 |
| Lichens and mosses | 0.0 | 0.0 |
| Bareland | 4.7 | 70.5 |
| Built-up | 75.8 | 80.0 |
| Water | 73.6 | 88.7 |
| Permanent Ice and Snow | 92.2 | 30.4 |
| Forest | 91.9 | 71.6 |

**Figure 33: Spatial pattern of agreement for 2022, showing consistent geographical patterns with 2000. The stable spatial distribution of agreement and disagreement indicates landscape-driven rather than temporal causes for inter-dataset discrepancies.**

The accuracy assessment results have implications for the use of these datasets in land cover monitoring applications [65]. The choice between datasets should consider specific application requirements. High-resolution GLC_FCS30D data is preferable for heterogeneous landscapes and fine-scale changes, while medium-resolution ESA CCI LC data is suitable for broad-scale trend analysis and long-term monitoring [66]. Applications should prioritize use of classes with high agreement (forest, cropland, water) while interpreting results from low-agreement classes with appropriate caution [44]. The observed inter-dataset discrepancies suggest that change detection results should be interpreted with awareness of inherent classification uncertainties [18]. Combining both datasets may provide complementary insights, using high-resolution data for detailed local analysis and medium-resolution data for consistent long-term monitoring [6].

Several methodological factors influenced the accuracy assessment results. The 10-fold difference in spatial resolution (30 m vs. 300 m) fundamentally limited the comparison, as many landscape features detectable at high resolution become mixed pixels at medium resolution [44]. Inherent differences in classification approaches between the two

datasets contributed to poor results, particularly for vegetation and transitional classes [50]. The majority filter resampling of high-resolution data may have introduced artifacts that affected the accuracy assessment, particularly in heterogeneous landscapes [42]. The assessment used one dataset as a reference for the other, rather than independent validation data, which may have influenced accuracy metrics [55].

The accuracy assessment revealed substantial but variable agreement between the GLC_FCS30D and ESA CCI LC datasets across Italy. The consistent ∼79% overall agreement suggests general consistency in land cover mapping between the two products, while the significant class-wise variations highlight important differences in how specific land cover types are conceptualized and detected [13, 53]. The stable accuracy patterns over time indicate robust performance of both classification systems, though persistent confusion among vegetation classes and for non-abundant land cover types suggests fundamental challenges in medium-resolution land cover mapping. These results provide guidance for researchers and practitioners in selecting and interpreting land cover data for environmental monitoring and policy applications in Italy. The assessment indicates the importance of understanding dataset characteristics and limitations when conducting land change analysis, particularly in heterogeneous Mediterranean landscapes where mixed pixels and class transitions present challenges for remote sensing classification [57, 65].

## 1.13. Land Cover Change Prediction

This section presents a predictive analysis of land cover changes in Italy from 2022 to 2032 using an enhanced Markov Chain approach integrated with realistic environmental constraints. This methodology leverages historical change patterns (last 12 years of the the time series) identified in previous analyses to generate spatially explicit and physically plausible projections of Italy's evolving landscape.

The analysis of historical land cover changes between 2010 and 2022 revealed significant transformation patterns across Italy, with particularly strong change signals in peri-urban areas, agricultural transition zones, and regions affected by climate change impacts [67]. These patterns informed both transition probability calculations and constraint parameters applied in the predictive modeling framework. The constrained Markov model generated spatially explicit predictions of land cover distribution for 2032, demonstrating both quantitative and spatial changes across Italy that integrate historical trends with realistic constraints based on geographical and policy considerations [68].




**Figure 34: Spatial distribution of land cover changes between 2010 and 2022. Areas in red indicate pixels that changed land cover class during this period, providing the empirical basis for transition probability calculations.**

**Table 13: Realistic constraints applied to Markov chain predictions based on Italian geographical realities and policy considerations**

| Land Cover Class | Max Area ($km^2$) | Min Area ($km^2$) | Annual Change Rate (%) |
|---|---|---|---|
| Built-up | 20,000 | 16,000 | +1.5 |
| Permanent ice and Snow | 800 | - | -2.0 |
| Water | 4,000 | - | +0.1 |
| Wetland | 1,000 | - | -1.0 |
| Bareland | 800 | - | -2.0 |
| Forest | - | 100,000 | -0.2 |
| Cropland | - | 140,000 | -0.3 |
| Grassland | - | 13,000 | -0.5 |
| Shrubland | - | - | +0.5 |
| Lichens and mosses | - | - | 0.0 |

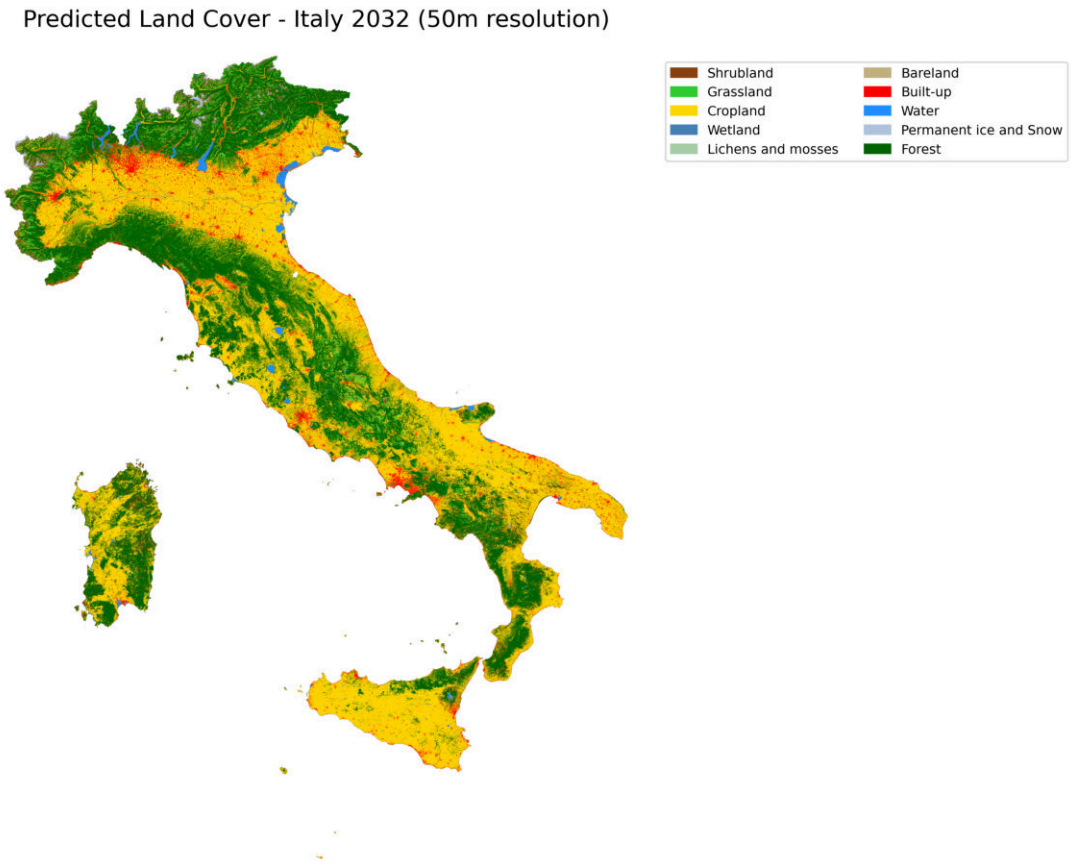



**Figure 35: Predicted land cover distribution for 2032 based on constrained Markov chain modeling, showing the integrated impact of historical trends and realistic constraints.**

Quantitative predictions revealed several significant changes across land cover classes (Table 14), reflecting the combined influence of historical transition patterns and applied constraints [69, 70]. The prediction identified several crucial land change trends aligning with both historical patterns and realistic constraints. Urban expansion manifested through a projected increase of 900.14 km$^2$ (+5.26%) in built-up areas, reflecting continued urbanization and infrastructure development concentrated in peri-urban areas and coastal regions, with growth rates constrained to realistic levels based on Italian urban planning policies [71]. Agricultural adjustments showed a moderate decline of 594.44 km$^2$ (-0.41%) in cropland, indicating continued consolidation and abandonment of marginal agricultural lands while maintaining stability in productive agricultural areas, aligning with historical patterns observed in the transition analysis [72].

**Table 14: Predicted land cover changes in Italy (2022–2032)**

| Land Cover Class | 2022 Area (km$^2$) | 2032 Area (km$^2$) | Change (km$^2$) | Change (%) |
|---|---|---|---|---|
| Shrubland | 11,163.78 | 11,866.15 | +702.37 | +6.29 |
| Grassland | 14,050.97 | 13,835.67 | -215.30 | -1.53 |
| Cropland | 144,901.97 | 144,307.52 | -594.44 | -0.41 |
| Wetland | 816.81 | 805.98 | -10.83 | -1.33 |
| Lichens and mosses | 133.88 | 138.35 | +4.47 | +3.34 |
| Bareland | 641.41 | 464.78 | -176.63 | -27.54 |
| Built-up | 17,120.87 | 18,021.01 | +900.14 | +5.26 |
| Water | 3,252.38 | 3,209.20 | -43.18 | -1.33 |
| Permanent ice and Snow | 997.82 | 807.03 | -190.79 | -19.12 |
| Forest | 108,316.17 | 107,940.36 | -375.81 | -0.35 |

Forest areas demonstrated minimal change (-375.81 km$^2$, -0.35%), suggesting effective conservation policies and sustainable management practices across Italy's diverse forest ecosystems, with minimum area constraints ensuring protection of this vital ecosystem [73]. The significant declines in permanent ice and snow areas (-190.79 km$^2$, -19.12%) is consistent with ongoing climate change impacts on alpine glaciers and snowpack, reflecting the accelerating impact of climate trends on sensitive cryospheric environments [74, 75]. Natural vegetation dynamics included substantial shrubland expansion (+702.37 km$^2$, +6.29%) likely due to abandonment of marginal agricultural lands and successional processes in previously disturbed areas, representing a natural response to agricultural abandonment patterns [76], alongside dramatic bareland reduction (-176.63 km$^2$, -27.54%) indicating successful land reclamation efforts and natural vegetation recovery.

Spatial predictions revealed distinct geographical patterns of land cover change aligning with both historical trends and geographical constraints [77]. Urban growth patterns showed concentration around major cities and along transportation corridors [78], with significant tourism-related development in coastal areas particularly along the Adriatic and Mediterranean coastlines [79], and limited urban growth in protected areas and mountainous regions due to planning restrictions [80]. Agricultural changes exhibited pronounced decline in marginal hilly areas and southern regions with lower agricultural productivity [81], while prime agricultural areas in the Po Valley and other lowland regions remained stable [82], with some agricultural intensification patterns visible in areas with irrigation infrastructure development [83].

Natural vegetation dynamics maintained forest stability across the Apennines and Alpine regions, reflecting effective conservation measures ,with shrubland expansion evident in abandoned agricultural areas particularly in central and southern Italy and grassland changes showing regional variations with some declines in managed grasslands but stability in natural grasslands [84]. Climate-sensitive changes demonstrated pronounced glacier retreat in the Alps, particularly in western and central sectors [85], water body fluctuations with some reservoir expansion but also drought impacts in southern regions [86], and wetland changes reflecting both conservation efforts and climate-related hydrological changes [87].

The predictive analysis integrated accuracy assessment findings through several mechanisms: leveraging historical transition patterns by incorporating change patterns identified between 2010 and 2022, ensuring projections remained grounded in empirically observed land cover dynamics with the substantial change area of 19,536 $km^2$ providing robust foundation for calculating realistic transition probabilities addressing dataset-specific characteristics by applying constraints that account for systematic biases and limitations identified in the accuracy assessment, particularly through conservative rates of change for low-agreement classes reflecting higher uncertainty associated with these classifications; and incorporating realistic change rates informed by historical accuracy patterns to ensure projected changes remained within physically plausible limits, mitigating potential for unrealistic extrapolation of historical trends.

The prediction model incorporated multiple validation measures to ensure reliability, building upon the methodological framework established in the accuracy assessment [88]. Area conservation was maintained through total consistency (301,396 $km^2$), validating spatial integrity of predictions. Historical validation demonstrated good correspondence between the 2010–2022 transition matrix and observed changes during this period. All predictions remained within physically realistic constraints based on Italian geograph-

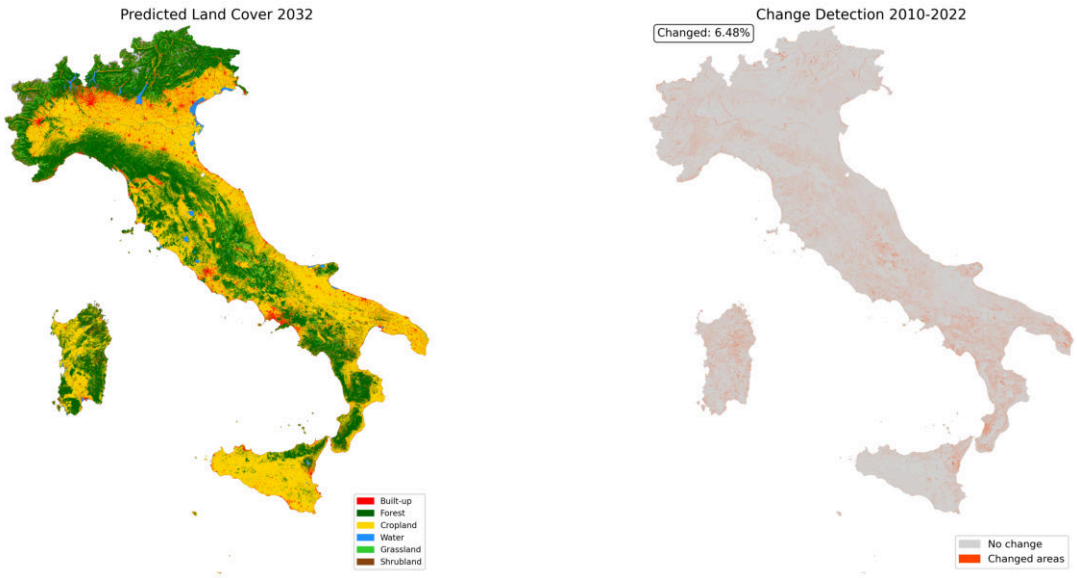



**Figure 36: Comprehensive visualization of predicted land cover changes integrating accuracy assessment findings**

ical realities, and uncertainty was acknowledged through conservative change rates in transition probabilities and constraint parameters.

The predictive analysis yields significant implications for Italian land management and policy. Projected urban expansion necessitates sustainable development strategies, particularly in peri-urban and coastal zones [89] Agricultural decline, especially on marginal lands, warrants targeted support programs and land consolidation initiatives. While forest stability suggests effective conservation policies, ongoing monitoring remains crucial. The dramatic cryospheric changes highlight an urgent need for climate adaptation in alpine regions. Shrubland expansion presents dual challenges for fire management and opportunities for biodiversity conservation, requiring integrated approaches.

Several limitations must be acknowledged within this predictive approach. The model's stationarity assumption may not hold under rapidly changing socioeconomic or climatic conditions [90]. Spatial homogeneity in transition probabilities potentially overlooks regional variations, while the exclusion of explicit socioeconomic drivers, policy changes, and climate impacts may limit realism [91]. Projections remain constrained by inherent dataset accuracy and resolution limitations identified in the accuracy assessment. Future research should address these gaps through several avenues: integration of spatially explicit transition probabilities based on regional characteristics incorporation of socioeconomic and climate scenarios to explore alternative futures development of hybrid modeling approaches combining statistical methods with process-based models and enhanced validation using independent data sources and participatory approaches [92].

## 1.14. Results Discussion

This section interprets the findings of the study in the broader context of land cover change science, focusing on the implications of observed trends, methodological insights from the multi-dataset analysis, and validation of the predictive model. The discussion synthesizes results relative to the research objectives and situates them within the existing literature.

The analysis reveals a dynamic landscape in Italy, shaped by interacting processes of urbanization, agricultural transformation, and ecological succession. Integrating multi-resolution datasets across temporal scales provides a nuanced perspective that goes beyond net change statistics to uncover the underlying transitions and their drivers.

Italy's agricultural landscape exhibits complex transformations. Net declines in cropland and increases in shrubland and forest confirm widespread abandonment in line with Mediterranean land change literature [9, 93], particularly in hilly and mountainous regions of the Apennines and the islands. Simultaneously, high-resolution data reveal areas of agricultural intensification and recultivation, including forest-to-cropland and shrubland-to-cropland conversions. These countervailing trends produce a *mosaic landscape of retreat and renewal*, where abandonment in marginal areas coexists with intensified cultivation in fertile plains such as the Po Valley, with implications for regional food security, biodiversity, and rural economies.

Urban expansion emerges as the clearest trend, with built-up areas more than doubling over the study period. This growth, averaging over 200 $km^2$ per year, aligns with national soil consumption reports [15, 94] and reflects demographic pressures, economic activity, and infrastructure development. Expansion is concentrated around major metropolitan areas and coastal zones, highlighting spatial polarization of economic activity and associated risks such as habitat fragmentation and loss of prime agricultural land. Notably, 30 m resolution data capture significant **land recycling** (Built-up $\rightarrow$ Cropland transitions), revealing processes of brownfield redevelopment, de-sealing, and urban infill. While currently overshadowed by net expansion, these processes indicate emerging cycles of urban land reuse that could support more sustainable urban planning.

Italy's forest cover shows near-neutral net change, but this masks significant gross fluxes between forest, shrubland, and cropland. Losses from logging, wildfires, and pests, exacerbated by climate change, are counterbalanced by afforestation and passive regrowth on abandoned farmland [95]. The projections to 2032 suggest this balance may persist in the short term but remain sensitive to climatic or policy shocks.

Methodologically, spatial resolution strongly influences the detection and interpretation of land change. The tenfold difference between datasets corresponded to a 6.25-fold difference in detected change. Coarse-resolution (300 m) data captured large-scale, unidirectional processes like urban expansion, whereas high-resolution (30 m) data were essential for detecting fine-grained, bidirectional transitions, such as cropland-shrubland cycles or subtle urban infill. For heterogeneous landscapes and small-scale land ownership, high-resolution monitoring is critical to accurately quantify land dynamics and inform management strategies.

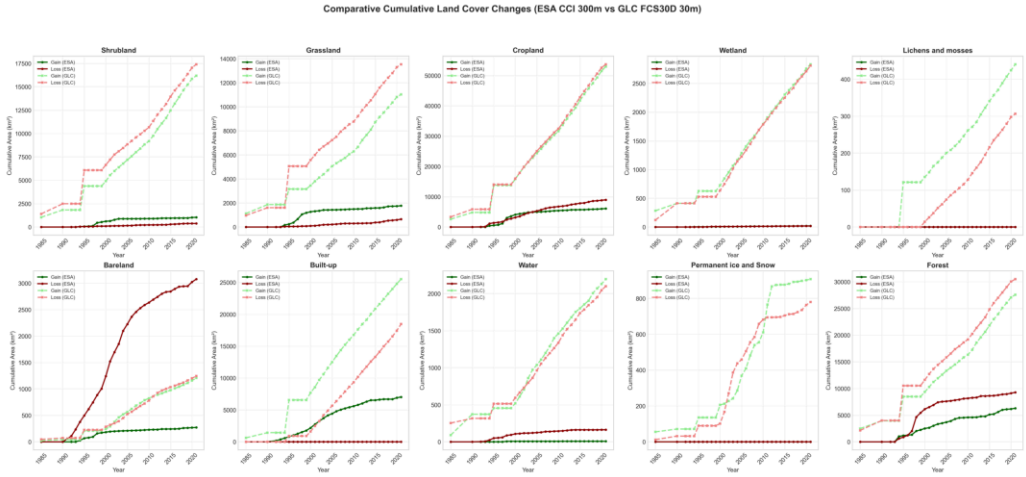



**Figure 37: Comparative cumulative land cover change (ESA CCI 300m vs. GLC FCS30D 30m), illustrating the impact of spatial resolution on change detection magnitude.**

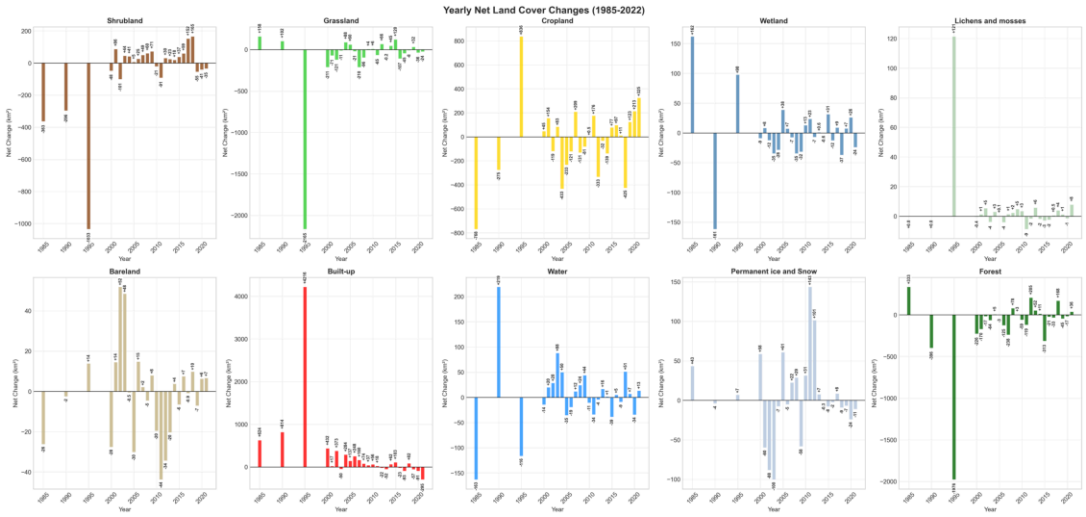



**Figure 38: Yearly net land cover change (1985–2022) showing volatility and key trends across different epochs.**

Climate change emerges as a current driver reshaping the Italian landscape. Observed losses of permanent ice and snow in the Alps are projected to continue, with a further 19% reduction by 2032, corroborating national-scale trends reported in previous studies. Impacts extend beyond alpine zones, affecting water availability for agriculture, urban centers, river ecosystems, and hydroelectric production, highlighting the need for adaptive water and land management.

Land change drivers vary regionally. The Po Valley in the North is marked by economic growth, urban expansion, and agricultural intensification. Central Italy shows pronounced agricultural abandonment and ecological succession. The South and islands display mixed patterns, including ongoing land degradation, pockets of modernization, and climate vulnerability. These variations necessitate differentiated policy responses, from managing soil sealing and intensification in the North, to supporting sustainable successional landscapes in the Centre, and addressing land degradation and climate risk in the South.

Overall, the findings reveal a complex Italian landscape undergoing accelerated transition, shaped by economic, demographic, policy, and climatic factors. The study highlights the value of high-resolution, multi-temporal monitoring and the need for regionally tailored policies to support sustainable land management and conservation.

## 1.15. Comparison with Previous Studies

This study contributes to the broader literature on land use and land cover change (LUCC) in Italy by integrating high-resolution change detection, multi-dataset accuracy assessment, and predictive modeling. While earlier research has examined specific aspects of Italy's landscape dynamics, the present work provides a comprehensive, spatially explicit analysis that both confirms established trends and highlights new insights.

The observed expansion of built-up areas approximately +104% according to ESA CCI LC and +70% according to GLC_FCS30D over their respective periodsaligns with previous national-scale assessments and refines their quantitative estimates. These results corroborate reports of soil consumption from the Italian National Institute for Environmental Protection and Research (ISPRA) [15, 94]. Transition analysis using the fine-resolution GLC_FCS30D data further captures the complexity of urban dynamics, including notable land recycling (Built-up → Cropland), a phenomenon less visible in coarser analyses. Declines in cropland and increases in shrubland and forest in marginal areas reflect processes of agricultural abandonment and rural depopulation documented

across Southern Europe [96, 97]. Spatially explicit evidence for Italy strengthens prior regional case studies; for example, land abandonment and subsequent regrowth in the Apennines aligns with findings from Basilicata [93], while shrubland expansion in Sicily and Sardinia matches the woody plant encroachment reported by [9]. A key contribution of this study is the national-scale quantification of these transitions using a consistent methodology, facilitating direct comparison across diverse agro-ecological regions and providing a baseline for policy monitoring.

Although net forest cover appears relatively stable, underlying transitions between forest, shrubland, and cropland are significant. The detection of gross changes (e.g., Forest $\leftrightarrow$ Shrubland) complements the National Forest Inventory (INFC) [95], which primarily reports net forest area and biomass. High-resolution data allowed identification of fine-scale processes such as forest degradation and recovery that are often obscured in national statistics. Observed and projected losses of permanent ice and snow (19.1% by 2032) offer quantitative support for glacier retreat documented in the Alps, including studies of the Adamello glacier and inventories in the Lombardy Alps [98]. Integrating these localized observations into a national-scale perspective emphasizes the impacts of climate change on Italy's cryosphere and the importance of adaptive water management strategies.

A further contribution is the systematic inter-comparison of ESA CCI LC and GLC_FCS30 datasets over Italy. While global validation studies exist [6, 18], few focus on Italy's heterogeneous Mediterranean landscapes with such detailed class-wise accuracy assessment. The observed disagreements in complex vegetation classes (shrubland, grassland) highlight the need for caution when using global products for national reporting without local validation. The constrained Markov chain model applied here advances beyond retrospective analyses by projecting land cover change based on observed physical transitions. Unlike economic or agent-based models applied in regions such as Lombardy or the Veneto coast, this data-driven approach incorporates Italy-specific constraints to ensure projections remain within plausible biophysical limits, enhancing the relevance for near-term ecological forecasting and policy planning.

In summary, this study consolidates regional evidence into a coherent national analysis. It confirms major drivers of change, including urbanization and agricultural abandonment, and contributes by: 1) quantifying the complexity of transitions underlying net change figures, 2) providing a rigorous accuracy assessment of global datasets in the Italian context, and 3) offering spatially explicit projections that integrate historical trends with environmental constraints. These findings support evidence-based decision-making for balancing economic development, agricultural sustainability, and environmental con-

servation in Italy.

## 1.16. Methodological Strengths and Limitations

This study applied a multi-scale analytical framework to examine Italy's land cover dynamics from 1985 to 2022, with projections to 2032. While this approach provides valuable insights into landscape transformations, it is important to critically assess both its strengths and limitations to properly contextualize the findings and guide future research.

A key strength of the study lies in the complementary use of two global land cover datasets: ESA CCI LC (300 m) and GLC_FCS30D (30 m) [12, 14]. The integration of these datasets enabled a resolution-sensitive analysis that surpassed the constraints of single-source studies. The 300 m data offered a consistent long-term perspective suitable for identifying broad-scale trends, while the 30 m data captured fine-grained processes characteristic of Italys heterogeneous Mediterranean landscapes. Implementing the analysis within GRASS GIS, enhanced by custom Python scripts, ensured robust geospatial processing and methodological reproducibility. Furthermore, the use of per-pixel area calculations accounted for distortions in the geographic coordinate system, particularly relevant given Italys latitudinal extent, enhancing the reliability of areal estimates.

The Markov chain modeling approach was strengthened by the incorporation of environmental constraints. Domain-specific knowledge, including maximum and minimum area limits, realistic change rates from national statistics, and strict area conservation rules, ensured that projections remained physically plausible and data-driven. This constrained approach addressed a common limitation of pure statistical models, which may otherwise produce unrealistic extrapolations. Systematic inter-comparison of the datasets using established accuracy metrics provided further insights into their reliability within the Italian context [18], highlighting class-specific uncertainties often overlooked in broader validation studies.

Several limitations should also be acknowledged. Harmonizing datasets into common aggregate classes could not fully resolve differences in original classification schemes and definitions [6], especially for transitional vegetation and heterogeneous landscapes, leading to unavoidable uncertainty in change detection. The resolution difference between datasets (30 m vs. 300 m) constrained detectable change types: finer resolution captured small-scale processes such as urban infill and field-level agricultural transitions, while coarser resolution primarily reflected larger, contiguous changes. Consequently, dif-

ferences in detected change area reflect both true landscape processes and the inherent capabilities of each sensor system.

The Markov chain model assumed stationary transition probabilities throughout the projection period, an assumption that may not hold under rapidly changing climatic or socioeconomic conditions. While environmental constraints mitigated unrealistic projections, the model could not incorporate nonlinear thresholds or regime shifts that may arise under future climate pressures. Another limitation is the focus on pattern description rather than causal explanation: while the analysis identifies what changes occurred and where, it does not directly attribute them to specific socioeconomic, political, or environmental drivers. Accuracy assessment relied on inter-dataset comparison rather than higher-accuracy reference data, meaning estimates are relative to the chosen reference rather than representing absolute ground truth.

These strengths and limitations should be viewed as complementary rather than contradictory. The multi-dataset approach addresses limitations of single-source analyses, and the constrained modeling reduces risks associated with pure statistical extrapolation. Future research could enhance this framework by integrating very high-resolution imagery, incorporating socioeconomic and climate data to enable causal analysis, applying deep learning to improve classification of transitional classes, or combining statistical and process-based models. Participatory validation involving local knowledge could further improve assessments in complex agricultural and semi-natural landscapes. Despite these limitations, the framework provides a robust and replicable approach for national-scale land change assessment in Italy and offers guidance for similar studies in other Mediterranean regions with heterogeneous landscapes and comparable data challenges.

## 1.17. Implications for Land Management and Policy

The analysis of Italy's land cover dynamics from 1985 to 2022, with projections to 2032, highlights several implications for land management and policy. Urban expansion of over 7,000 $km^2$, with built-up areas projected to increase by 5.26% by 2032, points to the need for sustainable urban planning. Policies should prioritize brownfield redevelopment and urban infill over greenfield development, especially in peri-urban zones and coastal areas under intense tourism-related pressure.

Agricultural policy requires a nuanced, region-specific approach to address divergent trends of abandonment in marginal areas and intensification in productive regions. Findings suggest that reforms of the Common Agricultural Policy (CAP) should go beyond

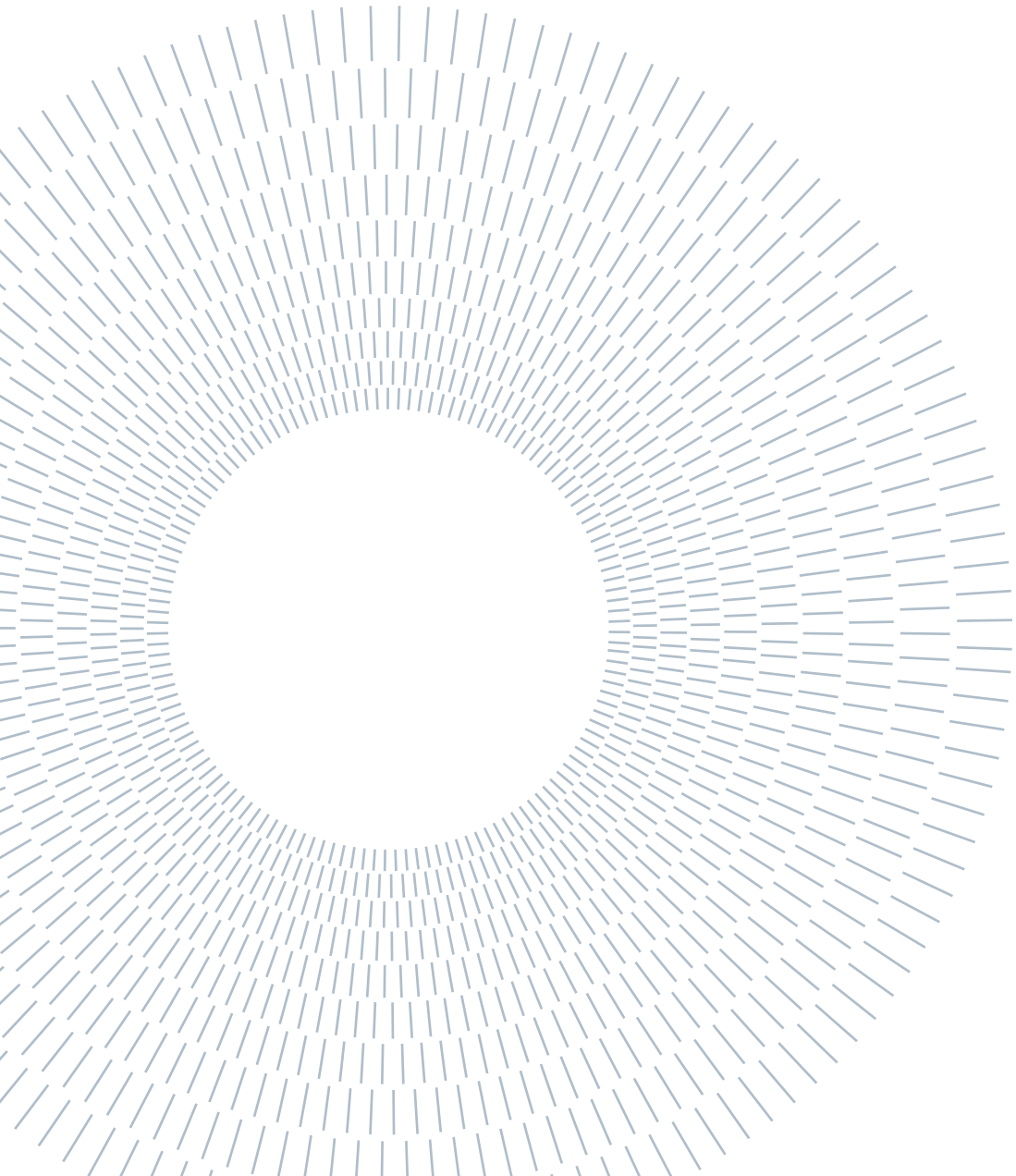
uniform measures, incorporating geographically differentiated incentives. These could support the maintenance of traditional landscapes in abandonment-prone regions while promoting sustainable intensification practices in productive areas, such as the Po Valley.

Projected climate-driven changes, including a 19.12% loss of permanent ice and snow by 2032, emphasize the need for integrated water resource management. Policies should anticipate future water demands for agriculture, urban consumption, and hydroelectric power, particularly in northern regions reliant on alpine water sources.

Forest management strategies should focus not only on area metrics but also on forest quality, species composition, and resilience to climate-related disturbances. The observed internal dynamics of forests despite net stability indicate that conservation efforts should prioritize ecological integrity over mere spatial extent.

The accuracy assessment highlights substantial uncertainties in monitoring semi-natural vegetation classes, underlining the importance of high-resolution, nationally-adapted land monitoring systems. Such systems are crucial for informed environmental governance, policy evaluation, and reporting under frameworks such as the EU Biodiversity Strategy and the Sustainable Development Goals.

Finally, regional patterns indicate the need for decentralized land governance. Northern Italy is characterized by urban and agricultural intensification, central regions by abandonment and ecological succession, and southern regions by mixed trajectories. Effective policy responses must recognize these diverse socio-ecological contexts, tailoring interventions to the specific needs and challenges of each region.



# Conclusion and Future Work

## 1.18. Summary of Findings

This study analyzed Italy's land cover dynamics from 1985 to 2022, revealing a landscape shaped by persistent urbanization, complex agricultural transitions, and dynamic forest changes. The integration of multi-resolution datasetsESA CCI LC (300 m) and GLC_FCS30D (30 m)provided a detailed perspective that would not be possible with single-source analysis. Results show substantial urban expansion, with increases ranging from approximately 70% (GLC_FCS30D) to 104% (ESA CCI LC) over their respective periods, , agricultural abandonment in marginal areas, and intricate forest dynamics where net stability conceals significant internal transitions. High-resolution data were essential for detecting fine-scale processes and bidirectional land changes, with the 30 m dataset capturing 6.25 times more change area than the 300 m dataset.

Projections using a constrained Markov chain model indicate that these trends are likely to continue through 2032, with ongoing urbanization, agricultural consolidation, and climate-driven losses of permanent ice and snow (19.1%). The comparison of datasets revealed significant disagreements for complex vegetation classes, underscoring the importance of local validation when using global products for national-scale applications.

## 1.19. Limitations

Several limitations of this research should be noted. Classification discrepancies between datasets, particularly for transitional vegetation classes, introduced uncertainty in detecting changes. The assumption of stationary transition probabilities in the Markov model may not hold under rapidly evolving climatic or socioeconomic conditions. Additionally, the analysis focused primarily on describing patterns rather than explaining causation, as socioeconomic and other driver variables were not integrated. Validation was limited to inter-dataset comparisons, rather than ground-truth verification, which constrains the interpretation of absolute accuracy.

## 1.20. Future Research Directions

Future research could build on this framework in several ways. Incorporating very high-resolution satellite imagery and ground-based validation would improve accuracy assessment, especially for challenging transitional classes. Integrating socioeconomic, demographic, and climate data would enable explanatory modeling to address causation rather than correlation. Hybrid modeling approaches that combine statistical methods with process-based representations of land change drivers could capture complex system dynamics more effectively. Participatory approaches that include local knowledge would further enhance understanding of fine-scale landscape changes and their societal implications.

Overall, this study provides a foundation for understanding land system dynamics in Italy and offers a replicable framework for similar analyses in other Mediterranean regions. The findings deliver evidence that can inform policymakers seeking to balance economic development, agricultural sustainability, and environmental conservation in a rapidly changing landscape.

# Appendix A: Supplementary Materials



```
"""
LAND COVER STATISTICS ANALYSIS SCRIPT - PART 1/2
Author: Ghulam Abbas Zafari
Thesis: Land Cover Dynamics and Transitions in
Italy: A Multi-Temporal Analysis Using
Global Land Cover Datasets
Institution: Milan Poiltechnic University

Part 1: Configuration, constants, and utility functions for land cover analysis.
This script processes annual land cover maps from 1985-2022 in GRASS GIS.
"""

import grass.script as gs
import pandas as pd
from tabulate import tabulate
from datetime import datetime
import os

# Get current timestamp for unique filenames
timestamp = datetime.now().strftime("%Y%m%d_%H%M%S")

# Class labels matching the land cover classification system
class_labels = {
    5: 'Shrubland',
    7: 'Grassland',
    8: 'Cropland',
    9: 'Wetland',
    11: 'Lichens and mosses',
    12: 'Bareland',
    13: 'Built-up',
    15: 'Water',
    16: 'Permanent ice and Snow',
    20: 'Forest',
    255: 'No data'
}

# Output directory configuration
output_dir = "glc_italy_30m_stats_output2"
os.makedirs(output_dir, exist_ok=True)

# Output filenames with timestamp
output_files = {
    'detailed': os.path.join(output_dir, f"glc_30m_detailed_stats_{timestamp}.csv"),
    'totals': os.path.join(output_dir, f"glc_30m_yearly_totals_{timestamp}.csv"),
    'summary': os.path.join(output_dir, f"glc_30m_summary_table_{timestamp}.csv")
}

# Target rasters (30m resolution) for analysis
TARGET_RASTERS = [
    "glc_1985@PERMANENT", "glc_1990@PERMANENT", "glc_1995@PERMANENT",
    "glc_2000@PERMANENT", "glc_2001@PERMANENT", "glc_2002@PERMANENT",
    "glc_2003@PERMANENT", "glc_2004@PERMANENT", "glc_2005@PERMANENT",
    "glc_2006@PERMANENT", "glc_2007@PERMANENT", "glc_2008@PERMANENT",
    "glc_2009@PERMANENT", "glc_2010@PERMANENT", "glc_2011@PERMANENT",
    "glc_2012@PERMANENT", "glc_2013@PERMANENT", "glc_2014@PERMANENT",
    "glc_2015@PERMANENT", "glc_2016@PERMANENT", "glc_2017@PERMANENT",
    "glc_2018@PERMANENT", "glc_2019@PERMANENT", "glc_2020@PERMANENT",
    "glc_2021@PERMANENT", "glc_2022@PERMANENT"
]

def format_number(num):
    """Format number with thousands separator and 2 decimal places"""
    return f"{num:,.2f}"

def format_percentage(num, total):
    """Calculate and format percentage"""
    return f"{(num/total)*100:.2f}%" if total > 0 else "0.00%"

def get_pixel_area(raster_name):
    """Get exact pixel area in square meters"""
    region = gs.parse_command('g.region', flags='pg', raster=raster_name.split('@')[0])
    nsres = float(region['nsres'])
    ewres = float(region['ewres'])
    return nsres * ewres

def generate_yearly_stats(raster_name):
    """Generate statistics for a single 30m raster"""
    try:
        # Get precise pixel area for this raster
        pixel_area_m2 = get_pixel_area(raster_name)

        # Validate pixel size (should be ~900 m² for 30m data)
        if not (890 < pixel_area_m2 < 910):
            print(f"^ Warning: Unexpected pixel area {pixel_area_m2:.2f} m² for {raster_name}")

        # Get statistics with actual areas in m²
        stats = gs.read_command('r.stats', flags='an', input=raster_name,
separator='space').strip().splitlines()

        if not stats:
            print(f"Warning: No statistics found for {raster_name}")
            return None, 0

        results = []
        total_area_m2 = 0
        year = raster_name.replace("glc_", "").replace("@PERMANENT", "")

        for line in stats:
            parts = line.split()
            if len(parts) < 2:
                continue

            cat = parts[0]
            area_m2 = float(parts[1])
            count = area_m2 / pixel_area_m2  # Calculate exact pixel count

            try:
                cat_int = int(cat)
                label = class_labels.get(cat_int, f'Class {cat}')
                area_km2 = area_m2 / 1_000_000

                results.append({
                    'class': cat_int,
                    'label': label,
                    'count': count,
                    'area_m2': area_m2,
                    'area_km2': area_km2,
                    'pixel_area_m2': pixel_area_m2
                })
                total_area_m2 += area_m2
            except ValueError:
                continue

        total_area_km2 = total_area_m2 / 1_000_000

        # Print summary
        print(f"\nStatistics for {year} (30m resolution):")
        print(f"Pixel size: {pixel_area_m2:.2f} m² | Total area: {format_number(total_area_km2)} km²")
        print(tabulate(
            [(r['class'], r['label'], f"{r['count']:,.0f}",
              format_number(r['area_km2']),
              format_percentage(r['area_km2'], total_area_km2))
             for r in sorted(results, key=lambda x: x['area_km2'], reverse=True)],
            headers=['Class', 'Label', 'Pixel Count', 'Area (km²)', '%'],
            tablefmt='simple'
        ))

        return results, total_area_km2

    except Exception as e:
        print(f"Error processing {raster_name}: {str(e)}")
        return None, 0
```

**Figure 39: Land Cover Statistics Analysis Script - Part 1: Configuration, constants, and utility functions for processing land cover data in GRASS GIS.**



```
"""
LAND COVER STATISTICS ANALYSIS SCRIPT - PART 2/2
Author: Ghulam Abbas Zafari
Thesis: Land Cover Dynamics and Transitions in
Italy: A Multi-Temporal Analysis Using
Global Land Cover Datasets
Institution: Milan Poiltechnic University

Part 2: Main processing functions and execution logic for land cover analysis.
This script computes area statistics for each land cover class and exports results.
"""

def generate_all_stats():
    """Generate statistics for all specified 30m rasters"""
    try:
        print(f"\nProcessing {len(TARGET_RASTERS)} land cover rasters (30m resolution)")

        all_data = []
        yearly_totals = []

        for raster_name in TARGET_RASTERS:
            stats, total_area = generate_yearly_stats(raster_name)
            if not stats:
                print(f"Warning: Could not process {raster_name}")
                continue

            year = raster_name.replace("glc_", "").replace("@PERMANENT", "")
            yearly_totals.append({
                'Year': year,
                'Total Area (km2)': total_area,
                'Formatted Area': format_number(total_area)
            })

            for item in stats:
                all_data.append({
                    'Year': year,
                    'Class': item['class'],
                    'Label': item['label'],
                    'Pixel_Count': item['count'],
                    'Area_km2': item['area_km2'],
                    'Area_m2': item['area_m2'],
                    'Pixel_Area_m2': item['pixel_area_m2'],
                    'Percentage': (item['area_km2']/total_area)*100 if total_area else 0
                })

        # Create DataFrames
        df_all = pd.DataFrame(all_data)
        df_totals = pd.DataFrame(yearly_totals)

        # Save to CSV
        df_all.to_csv(output_files['detailed'], index=False)
        df_totals.to_csv(output_files['totals'], index=False)

        print(f"\nSaved detailed statistics to '{output_files['detailed']}'")
        print(f"Saved yearly totals to '{output_files['totals']}'")

        return df_all, df_totals

    except Exception as e:
        print(f"Error in generate_all_stats: {str(e)}")
        return None, None

def main():
    """Main function to execute the land cover analysis"""
    print(f"\nStarting 30m land cover analysis at {timestamp}")

    # Verify projection
    proj_info = gs.read_command('g.proj', flags='p')
    if 'EPSG:3035' not in proj_info:
        print("⚠ Error: Projection must be EPSG:3035 (ETRS89-LAEA)")
        exit(1)

    detailed_df, totals_df = generate_all_stats()

    if detailed_df is not None and not detailed_df.empty:
        # Create summary table
        summary_df = detailed_df.pivot_table(
            index='Year',
            columns='Label',
            values='Area_km2',
            aggfunc='sum'
        ).fillna(0)

        # Order columns consistently
        column_order = [label for label in class_labels.values() if label in summary_df.columns]
        summary_df = summary_df[column_order]

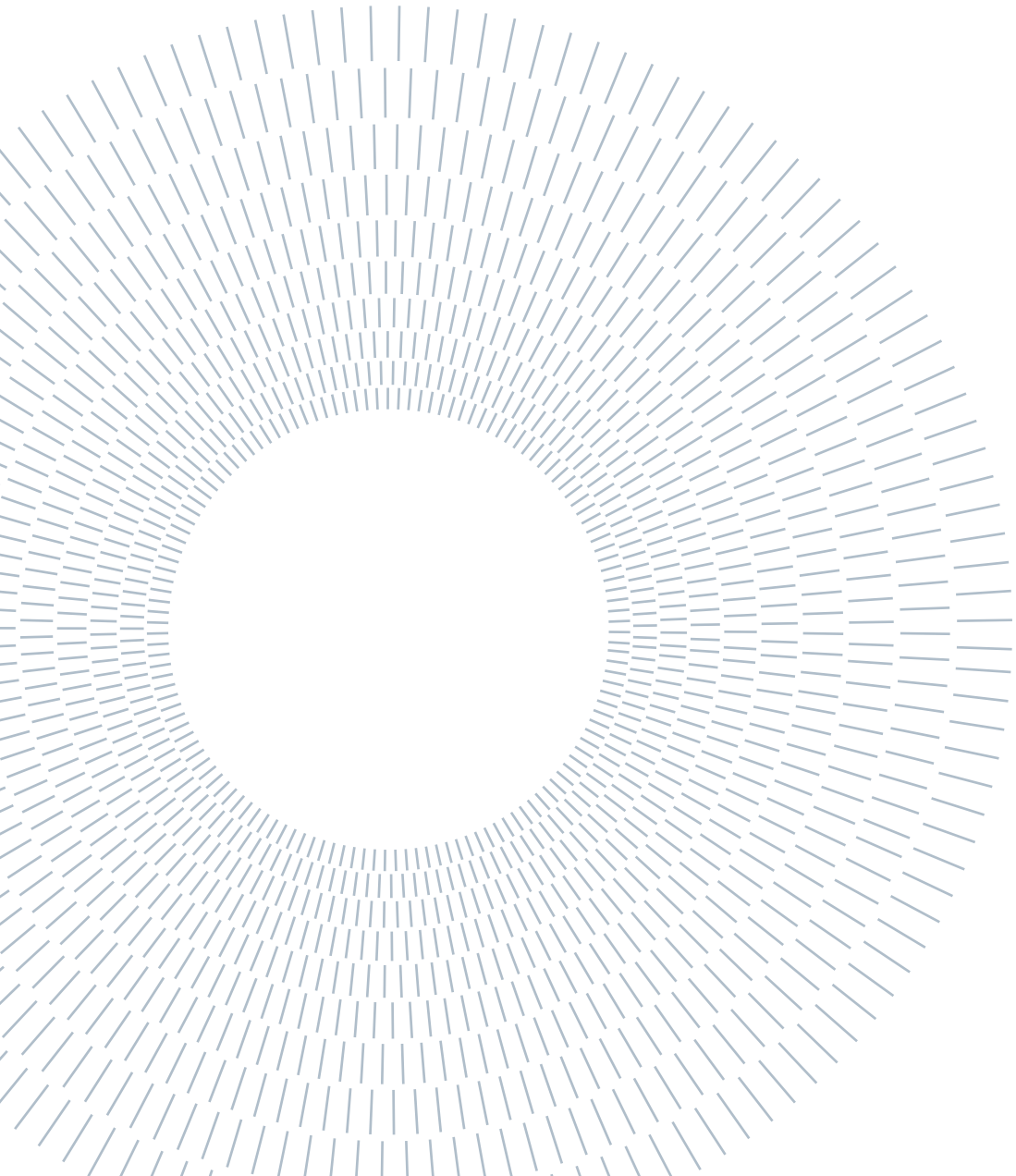
        # Save and display
        summary_df.to_csv(output_files['summary'])
        print(f"\nSummary statistics (30m resolution):")
        print(summary_df.to_string(float_format="%.2f"))
        print(f"\nSaved summary table to '{output_files['summary']}'")

        print("\nGenerated these output files:")
        for key, filename in output_files.items():
            print(f"- {filename}")

        # Validation check
        avg_pixel_size = detailed_df['Pixel_Area_m2'].mean()
        print(f"\nValidation: Average pixel size = {avg_pixel_size:.2f} m² (expected: ~900 m²)")
    else:
        print("\nNo data was processed - check for errors above")

if __name__ == "__main__":
    main()
```

**Figure 40: Land Cover Statistics Analysis Script - Part 2: Main processing functions and execution logic for computing area statistics and exporting results.**



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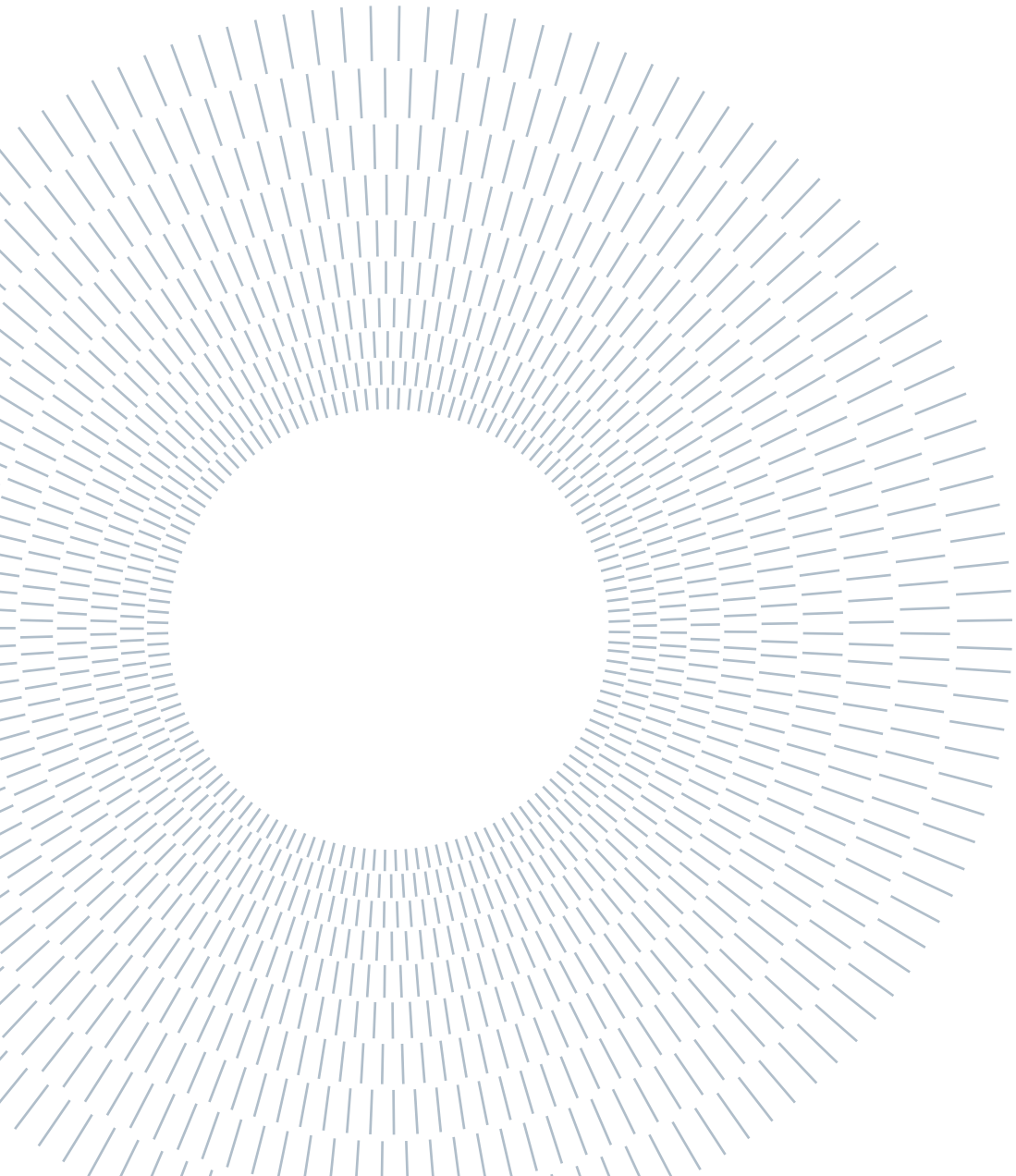
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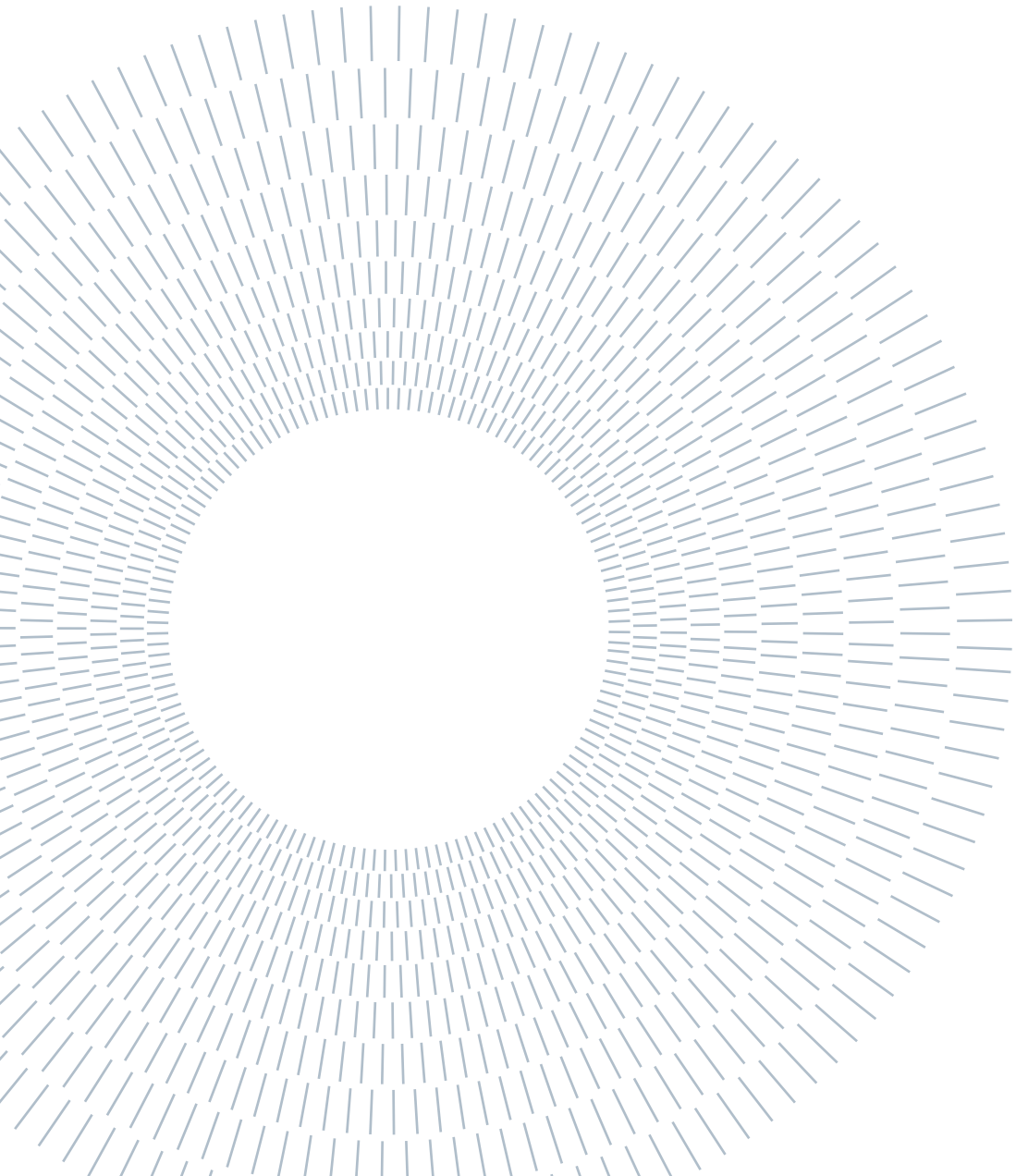
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